

Ethereum Order-Flow Alpha

Complete technical documentation: every stage, method, test and result
Binance USD-M ETHUSDT perpetual · 2023-03-01 → 2026-09-30 · replication on
SOLUSD

1. Raw data Binance archive 1,310 daily trade files, book depth, funding, OI SHA-256 verified	2. Quality control parse, ms/μs detect, sort, dedupe, gaps, invalid values, frozen book feed detection	3. 1-minute bars 1.88 bn trades → 1,886,400 bars with taker buy/sell flow + as-of book join	4. Features + targets 47 causal features 5 groups; forward returns 5/15/30/60 m leakage audit
5. Statistics HAC regressions, deciles + block bootstrap, events, Holm / BH	6. Walk-forward ML logit / RF / LightGBM 7 folds, purge + embargo, ablation A→E, tuning	7. Freeze → holdout methodology frozen + hash-logged, then one evaluation on 2025-10 → 2026-09	8. Cost-aware backtest next-bar VWAP, fees, slippage, funding, break-even cost, robustness, SOL

Final answer. Out-of-sample predictability is real and stable (holdout AUC **0.5436**, 95% CI 0.5394–0.5480), but the frozen strategy's break-even cost is **0.49 bps/side** versus a realistic **5.52 bps/side**: gross +67.5% (Sharpe 2.03), net **-99.6%**. **Statistically significant, economically untradable.**

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Part A — Overview

1. Executive summary

- **Data.** 1,877,097,488 Binance ETHUSD-perpetual aggregated trades (taker side recorded), 3,710,848 order-book percentage-band snapshots, funding and open interest; 3,976 files verified by SHA-256; aggregated to 1,886,400 one-minute bars.
- **Statistics.** Order flow explains the same minute's return (corr 0.467) but only weakly predicts the future: 1-minute OFI predicts a *reversal* of -0.15 to -0.24 bps per sd (HAC t -9.7 to -4.0 , Holm-significant).
- **Machine learning.** 43 walk-forward configurations; every binary classifier beats 0.5 AUC in all 7 folds (best ≈ 0.539). Primary model by pre-declared rule: rf_E_all_h15.
- **Holdout.** AUC 0.5436 [0.5394, 0.5480], ECE 0.0028, AUC > 0.5 in 12/12 months.
- **Economics.** gross Sharpe 2.03, break-even 0.49 bps/side vs 5.52 base cost; net -99.6%. Momentum and order-flow continuation rules lose money even before costs.
- **Incremental value of order flow.** Price-only models reach almost the full AUC; order flow and book bands add ≤ 0.002 AUC in development and nothing robust on the holdout.
- **Replication.** SOLUSD holdout AUC 0.5343; also untradable (break-even 0.89 bps/side).

2. Research question, hypotheses and design

Question. Does order-flow and market-microstructure information contain statistically significant predictive information about short-horizon ETH returns, and does it translate into economically meaningful alpha after costs?

Pre-registered hypotheses (labelled as hypotheses before any experiment) and their outcomes:

Hypothesis	Outcome
H1 OFI has a weak but detectable relation with very short-horizon returns	Confirmed (as reversal, not continuation)
H2 predictive performance only modestly above random	Confirmed (AUC ≈ 0.54)
H3 boosted trees beat linear models in some periods	Confirmed ($\approx +0.002$ to $+0.010$ AUC on holdout)
H4 performance deteriorates out of sample	Rejected (holdout AUC $\geq$ development AUC)
H5 costs substantially reduce or eliminate profitability	Confirmed: eliminated
H6 alpha varies across volatility regimes	Prediction similar across regimes; gross P&L; concentrated in mid/high vol
H7 ablation decides whether order flow adds information	Adds ≤ 0.002 AUC in development, nothing robust on holdout
H8 statistically significant but not economically tradable	Confirmed

Design principles

- Chronological splits only; a 12-month holdout fixed before any return was examined.
- Every feature uses information available at its decision time; verified automatically.
- All learned preprocessing fitted inside each training fold.
- Every choice (model, hyper-parameters, threshold, costs) made on development data by a pre-declared rule, then frozen and hash-logged.
- Statistical and economic significance reported separately; negative results reported as they are.

3. Project log: what was done, in order

When	What	Detail
Stage 1	Dataset selection	Probed Coinbase, Tardis (blocked/paid), Binance spot & futures; mapped bookDepth coverage for all 1,374 days; chose Binance USD-M ETHUSDT perpetual, 2023-03 → 2026-09
Stage 2	Acquisition + QC	Download with SHA-256 verification; switched monthly → daily trade files after one month needed 13.5 GB RAM; 1 transfer broke mid-stream → retry logic fixed, day re-processed
Stage 2	Book schema change	1,435,416 rows initially counted as invalid were Binance's new $\pm 0.2\%$ bands (from 2026-01-15); reclassified as schema extension, not used
Stage 2	Spread measurement	bookTicker on 3 development days: 1 tick in 99.5% of seconds; mean 0.046 bps → half-spread 0.023 bps
Stage 3	Features + targets	47 causal features, 4 horizons; first leakage audit; a too-strict equality check flagged 0 = 0 matches and was corrected to non-zero targets
Stage 4	EDA + statistics	HAC regressions, controls, deciles with bootstrap, event study, quarterly IC (development only)
Stage 5-7	Problem found	A model crashed on $\pm \text{inf}$ features → investigation found (a) a frozen order-book feed for a month, (b) 91 zero OI readings, (c) undefined ratios. Fixed in the pipeline; dataset rebuilt; audit re-run; all earlier model results discarded and re-run
Stage 5-7	Models	8-point LightGBM grid; ablation A-E × logit/RF/LightGBM; horizons; 3-class and regression targets; rule baselines
Stage 8	Dev backtest	Primary model rf_E_all_h15 by log-loss rule; $\delta = 0.05$ by base-cost Sharpe; every δ net-negative
Freeze	Methodology frozen	docs/final_methodology.md + frozen_parameters.json committed (71ca5ad) before any holdout access; RF stride consistency fixed before freezing
Holdout	Single evaluation	ETH final evaluation and SOL replication, both logged with the same methodology hash
Stage 9-10	Robustness + reports	Significance/snooping accounting, notebooks, research report, PDFs

Part B — Methods

4. Data acquisition

URL pattern: `https://data.binance.vision/data/futures/um/{daily|monthly}/{dataset}/{SYMBOL}/{SYMBOL}-{dataset}-{period}.zip` plus `.zip.CHECKSUM ("<sha256> <file>")`.

Dataset	Granularity	Fields
aggTrades	daily	agg_trade_id, price, quantity, first_trade_id, last_trade_id, transact_time, is_buyer_maker
bookDepth	daily	timestamp, percentage, depth, notional
fundingRate	monthly	calc_time, funding_interval_hours, last_funding_rate
metrics	daily	create_time, symbol, sum_open_interest, sum_open_interest_value, long/short ratios, taker ratio
bookTicker	daily (3 days)	update_id, best bid/ask price & qty, transaction_time, event_time

Procedure: fetch .CHECKSUM → stream zip while hashing → compare → write or reject (never use a mismatching file) → append to manifest (dataset, period, URL, bytes, expected/actual SHA-256, status, UTC time). Small files are fetched with 16 threads; trade days are processed by 4 parallel processes (download → verify → clean → aggregate → delete zip). Retries use exponential back-off for network errors, both on request and mid-transfer. Every step is idempotent and resumable.

5. Data cleaning and quality-control methods

Check	Method	Action
Timestamp unit	per file, min and max epoch must fall in one unit's range (ms 1e12-1e13, μ s 1e15-1e16, ns 1e18-1e19); otherwise fail	detect
Ordering	count rows with decreasing time / non-increasing id in raw order; then stable sort by (ts, id)	count + fix
Nulls / non-finite / bad side flag	any null field, non-finite price/qty, side not true/false	remove + count
Impossible values	price ≤ 0 , quantity ≤ 0 , first_trade_id > last_trade_id	remove + count
Out-of-period	timestamp outside the file's UTC day	remove + count
Duplicates	exact duplicate rows (keep one); then same agg_trade_id with different content (keep first)	remove + count
Sequence gaps	agg id gaps within and across files; underlying fill-id gaps	flag
Bad-tick candidates	price / minute median - 1 > 2%	flag, keep
Extreme bars	log return or log(high/low) > 3%; classified by reversal in the next bar and trade count	list, keep
Missing periods	runs of minutes with no trade on the complete grid	list, keep (price carried from the past)
Book: invalid / extra bands	null, non-finite, depth/notional ≤ 0 ; bands outside $\pm 1.5\%$ counted separately	remove + count
Book: completeness / monotonicity	10 bands per snapshot; cumulative depth non-decreasing in band width	count
Book: gaps, short days	inter-snapshot gaps > 90 s; day with < 90% of 2,880 snapshots	flag, never interpolate
Book: frozen feed	snapshot identical in all 20 values to its predecessor	flag; age measured from last change
Book: sign and time zone	implied band prices vs trade VWAP at clock shifts -8, -1, 0, +1, +8 h	verify
Funding / OI	duplicates, spacing vs interval, OI $\leq 0 \rightarrow$ missing, gaps > 5 min	count / fix

6. One-minute bars and the as-of order-book join

Bar t covers $[ts, ts + 1 \text{ min})$. Aggregates: open/high/low/close, volume, notional, VWAP = notional/volume, aggTrade count, underlying fill count, taker buy/sell volume, notional and trade counts (buy = is_buyer_maker false), mean/median/max/p90/p95 trade size within the bar, buy/sell volume from aggTrades $\geq \{1,5,10,25,50,100,250\}$ units, bad-tick count. The grid is complete; minutes without trades keep null prices (filled later only from the past).

Book join: each bar receives the last snapshot stamped strictly before its close (book timestamps have 1-second resolution, so $\leq ts + 59 \text{ s}$). book_age_s = bar close – time of the last snapshot whose values changed; book_stale = no snapshot or age $> 120 \text{ s}$. Stale books yield null book features.

7. Feature definitions (all 47; decision time = bar close; trailing windows only)

Feature	Definition
ret_k	$\log(P_t / P_{t-k})$, $k \in \{1,5,15,30,60\}$; P = last trade price at/before bar close (forward-filled backwards only)
vol_w	rolling std of 1-min log returns over $w \in \{15,60,240,1440\}$ minutes (vol_1440 requires ≥ 720 obs)
mom_240_riskadj	$\text{ret}_{240} / (\text{vol}_{240} \cdot \sqrt{240})$: 4-hour momentum in volatility units
dist_ma_w	$\log(P_t / \text{mean}(P \text{ over last } w \text{ minutes}))$, $w \in \{60,240\}$: distance from moving average
range_15	$\log(\text{max high} / \text{min low})$ over the last 15 minutes
dist_vwap_15	$\log(P_t / \text{VWAP of the last 15 minutes})$
hour_sin, hour_cos	sin / cos of the UTC time of day (intraday seasonality)
log_volume_1	$\log(\text{volume}_t)$
volume_z_1440	z-score of log volume against its trailing 1-day mean and sd
volume_ratio_w	volume over last w minutes / ($w \cdot$ trailing 1-day mean minute volume), $w \in \{15,60\}$
trade_intensity_15	$\log(1 + \text{mean number of aggTrades per minute over 15 minutes})$
trade_count_z_1440	z-score of $\log(1 + \text{trade count})$ against trailing 1 day
ofi_w	$(\sum \text{taker-buy vol} - \sum \text{taker-sell vol}) / (\sum \text{buy} + \sum \text{sell})$ over $w \in \{1,5,15,60\}$ minutes; null if no volume
trade_imb_w	$(\text{buy trades} - \text{sell trades}) / (\text{buy} + \text{sell trades})$, $w \in \{1,15\}$
signed_flow_15	$\sum(\text{buy} - \text{sell volume})$ over 15 min / ($15 \cdot$ trailing 1-day mean minute volume)
ofi_w_z_1440	trailing 1-day z-score of ofi_w, $w \in \{1,15\}$
avg_trade_size_15	$\log(\text{volume} / \text{trades})$ over 15 minutes
avg_trade_size_z_1440	trailing 1-day z-score of log average trade size
large_share_15	volume of aggTrades $\geq 10 \text{ ETH}$ / total volume, 15 minutes
large_ofi_15	imbalance of buy vs sell volume from aggTrades $\geq 10 \text{ ETH}$, 15 minutes
obi_kpct	$(\text{bid depth within } k\% - \text{ask depth within } k\%) / \text{sum}$, $k \in \{1,2,3,5\}$; null if the book is stale
obi_1pct_chg_15	$\text{obi_1pct}_t - \text{obi_1pct}_{t-15}$
obi_1pct_mean_15	15-minute mean of obi_1pct
depth_1pct_z_1440	trailing 1-day z-score of $\log(\text{bid} + \text{ask depth within } 1\%)$
funding_rate_last	last settled funding rate (settlement time $\leq$ decision time)
funding_rate_mean_3	mean of the last 3 settled funding rates ($\approx 24 \text{ h}$)
minutes_to_funding	minutes until the next scheduled funding settlement
oi_chg_w	log change of open interest over $w \in \{60,1440\}$ minutes, OI rows lagged $\geq 5 \text{ min}$

Ablation sets: **A_price** = 16 features; **B_price_volume** = 22 features; **C_plus_flow** = 35 features; **D_plus_book** = 42 features; **E_all** = 47 features.

Large-trade threshold rule: size bucket closest to the median per-minute 95th-percentile aggTrade size in the initial training window (2023-03..2024-02) $\rightarrow 10 \text{ ETH}$ (median p95 = 12.89 ETH); for SOL $\rightarrow 250 \text{ SOL}$.

8. Targets

Target	Definition
fwd_ret_h	$(P[t+h] - P[t]) / P[t]$, $h \in \{5,15,30,60\}$; null if the horizon passes the data end
up_h (primary, $h = 15$)	1 if fwd_ret_h > 0 else 0
cls3_h	UP if > +11 bps, DOWN if < -11 bps, else NEUTRAL (band = base round-trip cost, fixed a priori)
label_end_h	ts + h + 1 min: the last instant whose price the label uses (for purging)

9. Statistical methods

Method	Specification
Contemporaneous correlation	$\text{corr}(\text{OFI}_1, r_1)$: how much flow explains same-minute price change
Univariate predictive regression	$\text{fwd_ret_h (bps)} = a + b \cdot z(\text{signal}) + e$; Newey-West HAC SE, Bartlett kernel, 2h lags (overlap induces MA(h-1) errors)
Non-overlapping re-estimate	same regression on every h-th observation with HAC(1): checks the overlap correction
Controlled regression	add <code>ret_15</code> , <code>ret_60</code> , <code>vol_60</code> , <code>volume_z_1440</code> (standardised); HAC 2h lags
Rank correlation	Spearman correlation on every 5th observation
Decile analysis	mean forward return per signal decile; Spearman monotonicity of decile means
Top-bottom spread inference	moving-block bootstrap over UTC days (5-day blocks, 500 resamples): SE, 95% CI, normal p-value
Event study	events: <code>OFI_1</code> , <code>OFI_15</code> beyond their 95th/5th percentiles; mean cumulative return at 1/5/15/30/60 min with day-bootstrap CI; volatility response = post-event $\Sigma r $ / unconditional
Stability	quarterly Spearman IC of <code>ofi_15</code> , <code>obi_1pct</code> and <code>ret_15</code> with <code>fwd_ret_15</code>
Multiple testing	Holm (FWER) and Benjamini-Hochberg (FDR) within each family; Bonferroni for the holdout AUC across 43 configurations
Holdout AUC / Sharpe inference	5-day moving-block bootstrap over days on frozen predictions / daily P&L;
Deflated-Sharpe hurdle	expected max Sharpe of N zero-skill strategies $\approx \sqrt{(2 \ln N)} / \sqrt{\text{years}}$ with $N = 43 \times 5$

10. Machine-learning models and hyper-parameters

Model	Specification
Logistic regression	<code>Pipeline(SimpleImputer(median, add_indicator) → StandardScaler → LogisticRegression(L2, C = 0.1, max_iter 500))</code>
Random forest	<code>Pipeline(SimpleImputer → RandomForest({"n_estimators": 200, "min_samples_leaf": 500, "max_depth": 10}, max_features sqrt, seed 20231)); training stride 15</code>
LightGBM	selected <code>{"num_leaves": 15, "min_child_samples": 500, "n_estimators": 200}</code> ; fixed: <code>learning_rate 0.03, subsample 0.8, colsample 0.7, reg_lambda 10</code> ; native missing handling
LightGBM grid	<code>num_leaves {15, 63} × min_child_samples {500, 5000} × n_estimators {200, 600}</code> ; criterion: mean validation log loss
3-class / regression	LightGBM multiclass (macro OvR AUC, UP-vs-DOWN AUC excl. neutral) / LightGBM regression on <code>fwd_ret_15</code> in bps (MAE, RMSE, IC)
Rule baselines	prior-only; momentum = <code>sign(ret_15)</code> ; flow rule = <code>sign of ofi_15_z_1440 beyond ±1; ofi_1 as a score</code>

11. Validation scheme

fold	train start	train end (excl.)	valid start	valid end (excl.)
F1	2023-03-01	2024-02-29	2024-03-01	2024-06-01
F2	2023-03-01	2024-05-31	2024-06-01	2024-09-01
F3	2023-03-01	2024-08-31	2024-09-01	2024-12-01
F4	2023-03-01	2024-11-30	2024-12-01	2025-03-01
F5	2023-03-01	2025-02-28	2025-03-01	2025-06-01
F6	2023-03-01	2025-05-31	2025-06-01	2025-09-01
F7	2023-03-01	2025-08-31	2025-09-01	2025-10-01

Purge: training rows need `label_end ≤ train_end` (= valid start – 1-day embargo). Validation rows need `label_end ≤ valid end`, so no development label uses holdout prices. Training stride 5 minutes (RF 15). Holdout: same scheme with four 3-month blocks, re-fitting the frozen model before each block on all data up to block start – 1 day.

12. Signals, backtest equations, costs and metrics

Item	Definition			
Signal	$s_t = +1$ if $P(\text{up}) > 0.5 + \delta$; -1 if $P(\text{up}) < 0.5 - \delta$; 0 otherwise (NaN $\rightarrow$ 0)			
Net position	$W_t = (1/H) \sum_{j=0..H-1} s_{t-j}$, $H = 15$; $ W \leq 1$, no leverage			
Execution	orders from bar t fill at $e_{t+1} = \text{VWAP}$ of bar $t+1$			
Gross P&L;	$\text{gross}_t = W_t \cdot (e_{t+2} / e_{t+1} - 1)$			
Costs	$\text{cost}_t = W_t - W_{t-1} \cdot c$, $c = \text{fee} + \text{slippage} + \text{half spread}$ (per side)			
Funding	$\text{funding}_t = -W_t \cdot f$ if a settlement falls in W_t 's holding interval (longs pay $f > 0$)			
Net	$\text{net}_t = \text{gross}_t - \text{cost}_t + \text{funding}_t$; $\text{equity} = \Pi(1 + \text{net}_t)$			
Trade-level	each nonzero $s_t = \text{one trade}$, entry e_{t+1} , exit e_{t+1+H} , net of a full round trip: win rate, profit factor, average trade			
Break-even	$c^* = (\sum \text{gross} + \sum \text{funding}) / \sum \Delta W $ (per side)			
Risk metrics	daily UTC aggregation; Sharpe and Sortino $\times \sqrt{365}$; max drawdown on equity; Calmar = annual return / MDD ; turnover/day; exposure = mean $ W $			
Classification	AUC, log loss vs prior, Brier, accuracy/precision/recall/F1 at 0.5, ECE (10 equal-count bins), Spearman IC, tradable AUC vs $\text{VWAP}_{t+1} \rightarrow \text{VWAP}_{t+16}$ return			
scenario	fee (bps)	slippage (bps)	half spread (bps)	total per side (bps)
low	2.0	0.5	0.023	2.523
base	4.5	1.0	0.023	5.523
high	5.0	2.0	0.023	7.023

Fee source: Binance USD-M VIP0 maker 2 bps / taker 5 bps, 10% BNB discount. Half spread measured: 0.023 bps (mean spread 0.046 bps on 2023-06-14, 2023-11-15, 2024-03-13). Slippage is an assumption.

13. Holdout protocol and freeze

- All selections made on development out-of-fold data with pre-declared rules (model: lowest mean validation log loss on set E; δ : highest base-cost Sharpe).
- `scripts/freeze_methodology.py` writes `docs/final_methodology.md` and `frozen_parameters.json`; it refuses to run once the holdout access log exists.
- Holdout rows are reachable only through `validation.unlock_holdout()`, which requires the methodology file and appends {time, purpose, SHA-256} to `reports/holdout_access_log.jsonl`.
- Freeze committed in 71ca5ad; accesses: 2026-10-07T14:21:37+00:00 (final evaluation ETHUSD, f890b16b518e...); 2026-10-07T14:41:34+00:00 (replication evaluation SOLUSD, f890b16b518e...)

Part C — Tests and verification

14. Unit-test catalogue (30 tests, all passing)

Run with `pytest -q`. Tests use synthetic data with known answers plus a 1-hour byte-identical subset of the official archive (`data_sample/`).

file	test	what it verifies
Stage 2	<code>test_url_convention</code>	archive URL and .CHECKSUM URL are built exactly as Binance publishes them
Stage 2	<code>test_periods</code>	monthly/daily period generation, incl. leap day and year roll-over
Stage 2	<code>test_sample_checksums</code>	SHA-256 of the committed sample files matches SHA256SUMS
Stage 2	<code>test_detect_time_unit</code>	epoch unit detection for ms / μ s / ns timestamps (parametrised $\times 3$)
Stage 2	<code>test_detect_time_unit_mixed_raises</code>	a file mixing units raises instead of guessing
Stage 2	<code>test_clean_aggtrades_counts_and_policy</code>	removal counts for price ≤ 0 , qty ≤ 0 , out-of-day, exact and conflicting duplicates; bad tick flagged but kept; taker-side mapping
Stage 2	<code>test_unsorted_input_is_sorted</code>	out-of-order raw rows are counted and sorted chronologically
Stage 2	<code>test_aggregate_1min_values_and_grid</code>	OHLC, volume, buy/sell split, VWAP, fills, size buckets; empty minutes kept with null price; no-trade runs
Stage 2	<code>test_sample_end_to_end_conservation</code>	on real data: Σ bar volume = Σ trade volume, buy + sell = total, high $\geq$ low
Stage 2	<code>test_book_sample_wide_and_monotone</code>	book long $\rightarrow$ wide pivot; complete snapshots; cumulative depth monotone
Stage 2	<code>test_asof_book_has_no_lookahead</code>	a snapshot stamped exactly at the bar close is NOT used for that bar
Stage 2	<code>test_asof_book_flags_stale</code>	old snapshots flag the bar as stale
Stage 2	<code>test_asof_book_frozen_feed_is_stale</code>	repeated identical snapshots age from the last change and become stale
Research	<code>test_order_flow_and_imbalance_definitions</code>	OFI-1, OFI-15 (volume-aggregated) and book imbalance match their formulas; no padding of short history
Research	<code>test_imbalance_zero_denominator_is_null</code>	zero volume gives null imbalance, not 0
Research	<code>test_stale_book_gives_null_book_features</code>	stale book $\rightarrow$ null book features
Research	<code>test_rolling_features_are_trailing</code>	features on a prefix equal the full-sample features (no look-ahead); <code>ret_15</code> formula
Research	<code>test_forward_fill_only_backward</code>	missing prices are filled from the past only; execution price flagged
Research	<code>test_targets</code>	forward return, binary and 3-class labels, null tail, label end time
Research	<code>test_walk_forward_purge_embargo</code>	7 folds; train ends embargo before validation; purge of overlapping labels
Research	<code>test_holdout_guard</code>	holdout refused without the frozen methodology; access logged with its SHA-256
Research	<code>test_signal_rules</code>	probability, flow and momentum signal rules incl. NaN $\rightarrow$ flat
Research	<code>test_staggered_position</code>	net position = mean of the last H signals
Research	<code>test_costs_per_side</code>	fee + slippage + half spread arithmetic
Research	<code>test_backtest_pnl_execution_timing_and_costs</code>	P&L; earned from next-bar VWAP (not the signal bar); entry+exit cost; trade return; break-even formula
Research	<code>test_funding_charged_to_position_held_at_settlement</code>	funding hits the position held over the settlement; shorts receive positive funding
Research	<code>test_multiple_testing_adjustments</code>	Holm and Benjamini-Hochberg adjusted p-values
Research	<code>test_hac_recovers_slope</code>	HAC OLS recovers a known slope on simulated data

15. Leakage audit (scripts/leakage_audit.py, verbatim output)

Generated by `scripts/leakage_audit.py`. Every number below is computed by that script.

Overall: PASS - no leakage detected

PASS: Truncation invariance of all features

47 features, 12 random cut points in 2024-04..2024-06 (3 months of bars), 34,774,174 feature values compared. Max relative difference: 0. Features failing: none.

PASS: Book snapshots joined strictly before bar close

1,886,389 bars with a snapshot; violations (snapshot ts $\geq$ bar close): 0; minimum snapshot age at decision time: 1 s. Book features are set to null when the snapshot is stale (> 120 s); stale/missing book data are never interpolated.

PASS: Funding rate known at decision time

1,886,400 rows matched to a settled funding record; records settled after the decision time: 0. Only realised (settled) rates are used, never the next predicted rate.

PASS: Open interest lagged at least 5 minutes

Minimum lag between OI create_time and decision time: 5 min (rule: ≥ 5 min).

PASS: Targets are forward returns from the decision price

h=5: max |error| 0; last 5 rows null: True; label_end ok: True; h=15: max |error| 0; last 15 rows null: True; label_end ok: True; h=30: max |error| 0; last 30 rows null: True; label_end ok: True; h=60: max |error| 0; last 60 rows null: True; label_end ok: True. $P[t]$ is the last trade price at or before the close of bar t; empty minutes carry the last price forward (forward fill only ever looks backwards).

PASS: No feature mirrors a target

Largest |Pearson corr(feature, fwd_ret_h)| on development data (every 7th minute): 0.0372 (ret_5, h=5); next: 0.0294 (ret_1, h=5). Exact equality with a non-zero target: max share over all feature/horizon pairs 0.00e+00 (threshold 1e-4); flagged pairs: none.

PASS: Walk-forward purge / embargo / holdout separation

h=5: all 7 folds pass (train label end $\leq$ valid start - embargo; valid labels end inside block; train strictly before validation); h=15: all 7 folds pass (train label end $\leq$ valid start - embargo; valid labels end inside block; train strictly before validation); h=30: all 7 folds pass (train label end $\leq$ valid start - embargo; valid labels end inside block; train strictly before validation); h=60: all 7 folds pass (train label end $\leq$ valid start - embargo; valid labels end inside block; train strictly before validation). Development rows end at 2025-09-30 23:59:00+00:00; the development rows whose label reaches into the holdout are dropped from every development fit because validation requires label_end $\leq$ block end (the last block ends at the holdout start).

PASS: Learned preprocessing fitted inside each training fold

Logistic regression = Pipeline(SimpleImputer(median, add_indicator) -> StandardScaler -> LogisticRegression); Random forest = Pipeline(SimpleImputer -> RF); LightGBM handles missing values natively. Pipelines are re-fitted from scratch on each training fold, so no scaling or imputation parameter ever sees validation or holdout rows. Feature z-scores use trailing windows only (check 1).

Design rules enforced in code

- Decision time for bar t is its close (ts + 1 min); the trade executes at the VWAP of bar t+1 (backtest.py), never at the price that produced the signal.
- All rolling windows are trailing (rolling_* in Polars end at the current row); no shift(-k) appears in features.py (negative shifts exist only in targets.py).
- No backward fill anywhere; forward fill only for the last traded price.
- Regime cut-offs and the large-trade size threshold are estimated on the initial training window (2023-03-01 .. 2024-02-29) only.
- Holdout rows are only reachable through validation.unlock_holdout, which refuses without docs/final_methodology.md and logs every access with that file's SHA-256 (reports/holdout_access_log.jsonl).

Static scan of features.py for negative shifts / backward fills: none found.

16. Data-quality report (scripts/run_stage2.py, verbatim output)

Generated by `scripts/run_stage2.py` → `ethof.quality.build_report`. Period **2023-03-01** → **2026-09-30** (UTC). Every figure below is computed from the pipeline outputs.

1. Raw files

Verified files (SHA-256 matches the published `.CHECKSUM`): **3,976**.

dataset	status	n	GB
aggTrades	ok	1,310	25.31
bookDepth	ok	1,310	0.6448
bookTicker	ok	3	0.4971
fundingRate	ok	43	3.828e-05
metrics	ok	1,310	0.01575

2. Trades (aggTrades)

- Raw aggregated trades read: **1,877,097,488** (from 1,310 daily files)
- Clean trades used for bars: **1,877,097,488**
- Timestamp unit detected per file: `{'ms': 1310}`
- Out-of-order rows in raw file order: time decreasing 0; agg id not increasing 0 (all data are sorted by (ts, agg_trade_id) anyway)
- Removed as data errors: **0** (null/non-finite 0; price ≤ 0 0; qty ≤ 0 0; first_id > last_id 0; outside file's day 0; bad side flag 0)
- Duplicates: exact rows 0; conflicting same-id rows 0
- agg_trade_id gaps within files: 0 (0 ids); across consecutive files: 0 boundaries with a gap
- Underlying trade-id gaps (first_trade_id $\neq$ previous last_trade_id + 1): 3,789,115 gaps, 4,047,288 ids missing = 0.2152% relative to agg-trade count; overlaps 0. Flagged, not repaired: the archive omits these fills from aggTrades (see §8).
- Bad-tick candidates ($> 2\%$ from the minute's median price): **333,456** trades in 69 bars (flagged, kept). 63 of these 69 bars are also in the extreme-move list (§3): the flags come from crash/squeeze minutes whose intra-minute range exceeds 2 %, not from isolated bad prints, so nothing is removed.

3. One-minute bars

- Bars written: **1,886,400**; expected minutes in period: 1,886,400; duplicate timestamps: 0; sorted: True
- Bars with ≥ 1 trade: 1,886,347; **minutes without any trade: 53** (0.0028%) in 4 runs; runs ≥ 2 min: 3
- OHLC consistency violations (high < low, open/close outside [low, high]): 0

Longest no-trade runs (`docs/stage2/no_trade_minute_runs.csv`):

start	end	minutes
2023-09-12 08:34:00+00:00	2023-09-12 08:52:00+00:00	19
2025-08-29 06:18:00+00:00	2025-08-29 06:36:00+00:00	19
2024-10-28 16:21:00+00:00	2024-10-28 16:34:00+00:00	14
2024-10-28 16:36:00+00:00	2024-10-28 16:36:00+00:00	1

Extreme 1-minute bars ($|\log \text{return}|$ or $\log(\text{high}/\text{low}) > 3\%$), **83** bars (`docs/stage2/extreme_1m_bars.csv`):

ts	open	high	low	close	n_trades	logret	range_rel	classification
2023-03-12 17:26:00+00:00	1,484	1,535	1,483	1,514	36,657	0.02011	0.03448	sustained move (market event)
2023-03-24 11:47:00+00:00	1,780	1,781	1,720	1,739	27,697	-0.02329	0.03492	sustained move (market event)
2023-04-14 00:36:00+00:00	2,026	2,110	2,025	2,101	40,794	0.03629	0.04107	sustained move (market event)
2023-04-19 08:22:00+00:00	2,014	2,017	1,948	1,983	35,355	-0.01553	0.03439	sustained move (market event)

ts	open	high	low	close	n_trades	logret	range_rel	classification
2023-08-17 21:41:00+00:00	1,704	1,705	1,631	1,656	16,558	-0.02897	0.0441	sustained move (market event)
2023-08-17 21:42:00+00:00	1,645	1,656	1,600	1,600	17,569	-0.03398	0.03454	sustained move (market event)
2023-08-17 21:44:00+00:00	1,558	1,560	1,493	1,499	18,945	-0.03934	0.04407	reverted, high activity
2023-08-17 21:45:00+00:00	1,499	1,582	1,499	1,572	18,939	0.04759	0.05395	sustained move (market event)
2023-10-01 22:17:00+00:00	1,685	1,780	1,685	1,769	24,334	0.04826	0.05459	sustained move (market event)
2023-10-02 18:39:00+00:00	1,681	1,681	1,620	1,649	28,216	-0.01904	0.03695	sustained move (market event)
2023-12-11 02:12:00+00:00	2,282	2,289	2,155	2,156	36,715	-0.05725	0.06049	sustained move (market event)
2024-01-03 12:01:00+00:00	2,301	2,302	2,194	2,229	30,033	-0.03185	0.04796	sustained move (market event)
2024-01-03 12:08:00+00:00	2,240	2,241	2,157	2,197	23,169	-0.01938	0.03827	sustained move (market event)
2024-01-03 12:09:00+00:00	2,197	2,200	2,079	2,088	28,077	-0.05106	0.05635	sustained move (market event)
2024-02-28 17:31:00+00:00	3,382	3,383	3,153	3,213	38,459	-0.05128	0.07034	reverted, high activity
2024-02-28 17:32:00+00:00	3,215	3,350	3,212	3,330	30,672	0.03571	0.04194	sustained move (market event)
2024-03-05 19:56:00+00:00	3,347	3,358	3,227	3,268	29,955	-0.02407	0.03994	reverted, high activity
2024-03-05 19:57:00+00:00	3,269	3,346	3,205	3,346	24,338	0.02386	0.04318	sustained move (market event)
2024-04-12 17:28:00+00:00	3,370	3,371	3,244	3,319	31,676	-0.01539	0.03818	sustained move (market event)
2024-04-12 18:35:00+00:00	3,226	3,229	3,062	3,130	36,099	-0.03011	0.05309	sustained move (market event)
2024-04-13 19:56:00+00:00	3,082	3,083	2,971	3,016	29,887	-0.02156	0.03686	sustained move (market event)
2024-04-13 20:07:00+00:00	3,025	3,033	2,902	2,929	23,137	-0.03222	0.04429	sustained move (market event)
2024-04-13 20:09:00+00:00	2,918	2,925	2,792	2,865	30,161	-0.01854	0.04662	sustained move (market event)
2024-05-23 12:56:00+00:00	3,911	3,911	3,782	3,833	45,888	-0.02019	0.03363	sustained move (market event)
2024-05-23 20:00:00+00:00	3,794	3,794	3,525	3,651	70,477	-0.03836	0.07361	sustained move (market event)
2024-05-23 21:02:00+00:00	3,760	3,881	3,758	3,857	59,531	0.02547	0.03221	sustained move (market event)
2024-06-24 09:33:00+00:00	3,323	3,326	3,213	3,288	40,904	-0.01047	0.03455	sustained move (market event)
2024-07-05 02:59:00+00:00	2,975	2,977	2,882	2,927	36,350	-0.0163	0.03249	sustained move (market event)
2024-08-04 17:44:00+00:00	2,695	2,700	2,613	2,650	35,540	-0.0168	0.03272	sustained move (market event)
2024-08-05 00:57:00+00:00	2,556	2,556	2,475	2,512	36,819	-0.01746	0.03219	sustained move (market event)
2024-08-05 01:08:00+00:00	2,424	2,428	2,281	2,316	36,569	-0.04564	0.0626	sustained move (market event)
2024-08-05 01:10:00+00:00	2,341	2,350	2,112	2,158	32,664	-0.08136	0.1068	reverted, high activity
2024-08-05 01:11:00+00:00	2,158	2,267	2,153	2,249	33,506	0.04151	0.05194	sustained move (market event)
2024-08-05 01:12:00+00:00	2,250	2,289	2,219	2,285	35,626	0.01547	0.03098	reverted, high activity
2024-08-05 01:13:00+00:00	2,285	2,290	2,206	2,265	33,960	-0.008709	0.03739	reverted, high activity
2024-08-05 01:24:00+00:00	2,352	2,353	2,272	2,300	29,960	-0.02259	0.03479	sustained move (market event)

ts	open	high	low	close	n_trades	logret	range_rel	classification
2024-08-05 06:24:00+00:00	2,226	2,226	2,150	2,176	27,899	-0.02223	0.03474	sustained move (market event)
2024-08-27 21:48:00+00:00	2,482	2,496	2,400	2,443	26,970	-0.01583	0.03902	sustained move (market event)
2024-12-05 22:28:00+00:00	3,783	3,784	3,615	3,717	45,403	-0.01754	0.04561	sustained move (market event)
2024-12-09 21:04:00+00:00	3,648	3,660	3,523	3,547	35,586	-0.02803	0.03807	reverted, high activity

(43 more rows in the CSV)

4. Order book (bookDepth percentage bands)

Reminder: these are *cumulative resting quantity/notional within $\pm 1...5$ % of the reference price*, not level-by-level L2. No best bid/ask or spread is available from this source.

- Daily files expected: 1,310; present: 1,310
- Snapshots kept: **3,710,848** (expected $\approx 3,772,800$ at 2880/day = 98.358%)
- Raw rows 38,543,896; invalid removed 0; exact dups 0; conflicting dups 0; rows of extra bands not used (Binance added ± 0.2 % bands from 2026-01-15; schema extension, not errors) 1,435,416; incomplete snapshots ($\neq 10$ bands) 0; non-monotone cumulative depth 0
- Gaps between consecutive snapshots > 90 s: **9,515**
- Frozen feed:** 97,477 snapshots repeat all 20 band values of their predecessor (2.63%); the longest run is 94,730 snapshots (2025-04-16 10:31:05+00:00 $\rightarrow$ 2025-05-19 10:26:32+00:00). Book age is measured from the last *changed* snapshot, so these minutes are flagged `book_stale` (`docs/stage2/book_frozen_runs.csv`).
- Short/missing days (< 90% of 2880 snapshots), flagged `book_short_day` in the bars and **not interpolated:** 9

date	file_present	n_snapshots
2023-03-17	1	354
2023-03-24	1	2,523
2023-06-01	1	2,027
2024-06-12	1	2
2026-03-26	1	2,590
2026-04-09	1	2,575
2026-04-10	1	2,577
2026-04-16	1	2,576
2026-04-22	1	2,588

Longest snapshot gaps:

ts	prev	gap_s
2024-06-12 23:59:01+00:00	2024-06-11 23:59:31+00:00	86,370
2023-03-18 00:06:31+00:00	2023-03-17 07:49:31+00:00	58,620
2023-06-01 07:03:02+00:00	2023-05-31 23:57:03+00:00	25,559
2023-03-17 04:53:03+00:00	2023-03-16 23:57:03+00:00	17,760
2023-03-24 07:53:02+00:00	2023-03-24 04:59:03+00:00	10,439
2025-07-23 07:36:03+00:00	2025-07-23 06:06:32+00:00	5,371
2023-05-14 00:39:03+00:00	2023-05-13 23:56:31+00:00	2,552
2023-06-06 00:37:31+00:00	2023-06-05 23:55:05+00:00	2,546
2023-05-16 00:25:31+00:00	2023-05-15 23:53:32+00:00	1,919
2025-10-09 03:57:01+00:00	2025-10-09 03:33:33+00:00	1,408

Longest frozen-feed runs:

start	end	snapshots
2025-04-16 10:31:05+00:00	2025-05-19 10:26:32+00:00	94,730
2026-03-20 01:44:33+00:00	2026-03-20 04:26:32+00:00	296
2026-04-22 06:34:31+00:00	2026-04-22 09:25:00+00:00	284
2025-08-26 06:43:31+00:00	2025-08-26 08:17:01+00:00	171

start	end	snapshots
2025-12-01 12:11:12+00:00	2025-12-01 13:14:05+00:00	127
2025-05-19 10:28:01+00:00	2025-05-19 11:12:02+00:00	89
2025-08-26 11:24:32+00:00	2025-08-26 12:07:30+00:00	86
2025-07-23 07:38:03+00:00	2025-07-23 08:10:40+00:00	56

- Bars whose as-of snapshot is older than 120 s (book_stale): **58,081** (3.079%); bars with no snapshot at all: 11

Sign-convention / time-zone check. Average price of resting quantity within the 1% band (notional/depth) vs. the trade VWAP of the minute containing the snapshot, with the snapshot clock shifted by whole hours:

clock_shift_hours	n_matched	share_bid1_lt_vwap_lt_ask1	median_abs_bps_bandmid_vs_vwap
-8	3,709,804	0.2942	85.27
-1	3,710,644	0.6421	27.67
0	3,710,742	0.9406	3.144
1	3,710,627	0.643	27.86
8	3,709,786	0.2951	84.51

5. Funding rates

- n_rows: 3,930
- n_duplicates_removed: 0
- interval_hours_values: [8]
- n_irregular_spacing: 0
- rate_min: -0.0003653
- rate_max: 0.001017
- rate_mean: 6.605e-05
- first_ts: 2023-03-01 00:00:00.016000+00:00
- last_ts: 2026-09-30 16:00:00.002000+00:00

6. Open interest / metrics (5-minute)

- n_files: 1,310
- n_rows: 377,147
- n_unparseable_ts: 0
- n_exact_duplicates_removed: 0
- n_expected_rows: 377,280
- n_gaps_gt_5min: 4
- n_minutes_in_gaps: 665
- n_nonpositive_open_interest_set_null: 91
- n_null_open_interest_raw: 0

7. Outputs

output	files	MB
bars_1m	43	481.5
book_snapshots	1	439.5
funding	1	0.05571
metrics	1	16.4

Processed 1-minute bar schema (processed/bars_1m/*.parquet):

column	dtype
ts	Datetime(time_unit='us', time_zone='UTC')
open	Float64
high	Float64
low	Float64
close	Float64
volume	Float64
notional	Float64

column	dtype
n_trades	Int64
n_fills	Int64
buy_volume	Float64
sell_volume	Float64
buy_notional	Float64
sell_notional	Float64
buy_trades	Int64
sell_trades	Int64
trade_size_mean	Float64
trade_size_median	Float64
trade_size_max	Float64
trade_size_p90	Float64
trade_size_p95	Float64
last_trade_ts	Datetime(time_unit='us', time_zone='UTC')
n_bad_tick_flags	Int64
buy_volume_ge1	Float64
sell_volume_ge1	Float64
buy_volume_ge5	Float64
sell_volume_ge5	Float64
buy_volume_ge10	Float64
sell_volume_ge10	Float64
buy_volume_ge25	Float64
sell_volume_ge25	Float64
buy_volume_ge50	Float64
sell_volume_ge50	Float64
buy_volume_ge100	Float64
sell_volume_ge100	Float64
buy_volume_ge250	Float64
sell_volume_ge250	Float64
vwap	Float64
has_trades	Boolean
book_ts	Datetime(time_unit='us', time_zone='UTC')
bid_depth_1	Float64
bid_depth_2	Float64
bid_depth_3	Float64
bid_depth_4	Float64
bid_depth_5	Float64
ask_depth_1	Float64
ask_depth_2	Float64
ask_depth_3	Float64
ask_depth_4	Float64
ask_depth_5	Float64
bid_notional_1	Float64
bid_notional_2	Float64
bid_notional_3	Float64
bid_notional_4	Float64
bid_notional_5	Float64
ask_notional_1	Float64
ask_notional_2	Float64
ask_notional_3	Float64
ask_notional_4	Float64
ask_notional_5	Float64
book_frozen	Boolean
last_change_ts	Datetime(time_unit='us', time_zone='UTC')
book_age_s	Int64

column	dtype
book_stale	Boolean
book_short_day	Boolean
book_day_snapshots	Int64

Preview (last 3 traded minutes of the final month):

ts	op en	hig h	lo w	clo se	vol um e	n_tra des	buy_v olume	sell_vol ume	vw ap	bid_de pth_1	ask_de pth_1	book_ age_s
2026-09-30 23:57:00+00:00	2,6 84	2,6 84	2,6 84	2,6 84	310. 7	148	133.9	176.8	2,6 84	70,882	38,572	30
2026-09-30 23:58:00+00:00	2,6 84	2,6 84	2,6 84	2,6 84	283. 9	147	206.6	77.33	2,6 84	69,974	39,054	30
2026-09-30 23:59:00+00:00	2,6 84	2,6 85	2,6 84	2,6 85	963. 4	267	626.5	336.9	2,6 85	66,762	38,814	30

8. Data-quality issues relevant to the research

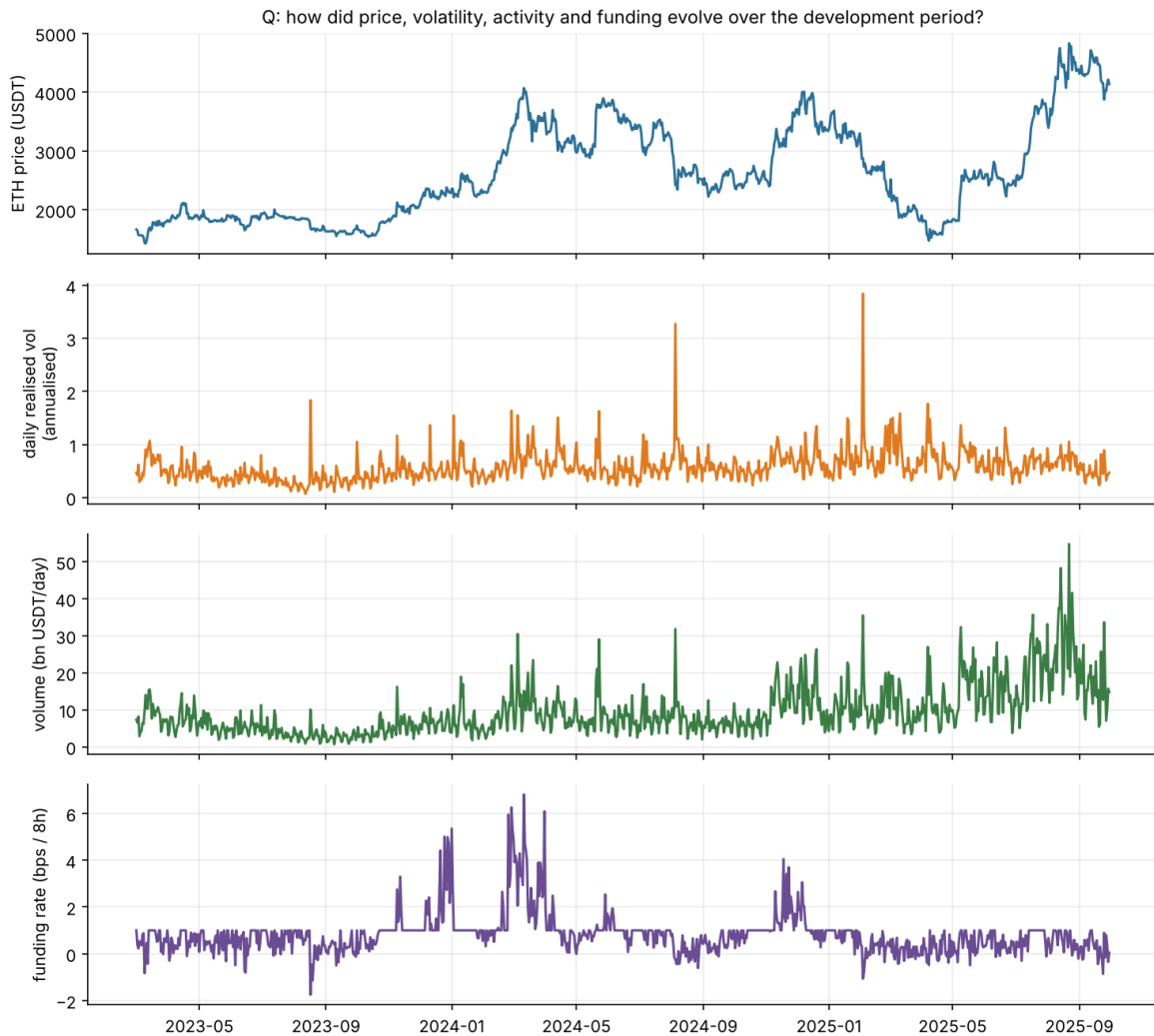
- Order book is coarse.** Percentage bands only; no best bid/ask, no spread, no queue information. Book features are null on 58,081 stale minutes (3.079%) and on the 9 short days listed in §4; nothing is interpolated.
- Frozen order-book feed:** 2025-04-16 10:31:05+00:00 → 2025-05-19 10:26:32+00:00 the archive repeats a single snapshot (94,730 identical snapshots; the daily files look complete but compress unusually well). Without value-change detection the book features would silently carry month-old information; they are null there instead. The open-interest series also contains 91 zero readings, set to missing.
- Schema change inside the holdout:** extra ± 0.2 % band rows appear from 2026-01-15. Not used, because they are unavailable in the development period.
- Extreme moves are real.** 83 one-minute bars move $> 3\%$; 0 meet the error-candidate rule (full reversal on < 20 trades). All are kept: they are liquidation cascades with tens of thousands of trades, i.e. exactly the events a strategy must survive.
- Trading halts / no-trade minutes:** 53 minutes in 4 runs. Prices are forward-filled for features (backward-looking only); positions cannot be traded there, so execution uses the last price and the minute is flagged.
- Fill-id gaps:** 4,047,288 underlying fill ids (0.216% of the agg-trade count) are absent from aggTrades, so volumes may be slightly understated. Whether the omitted fills are side-neutral cannot be verified from this archive.
- Single venue.** Binance perp flow only; cross-venue flow (Coinbase, Bybit, OKX, spot) is unobserved.

Sign-convention / time-zone figure data and frozen-feed runs

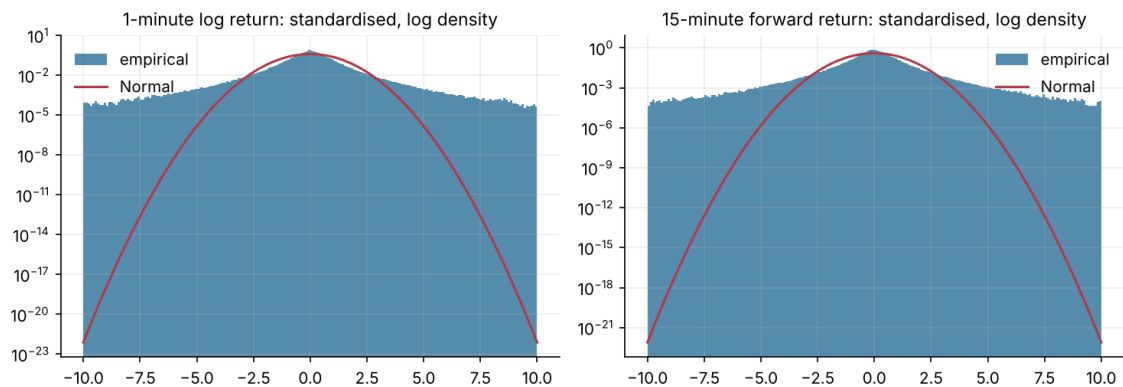
start	end	snapshots
2025-04-16T10:31:05.000000+0000	2025-05-19T10:26:32.000000+0000	94,730
2026-03-20T01:44:33.000000+0000	2026-03-20T04:26:32.000000+0000	296
2026-04-22T06:34:31.000000+0000	2026-04-22T09:25:00.000000+0000	284
2025-08-26T06:43:31.000000+0000	2025-08-26T08:17:01.000000+0000	171
2025-12-01T12:11:12.000000+0000	2025-12-01T13:14:05.000000+0000	127
2025-05-19T10:28:01.000000+0000	2025-05-19T11:12:02.000000+0000	89
2025-08-26T11:24:32.000000+0000	2025-08-26T12:07:30.000000+0000	86
2025-07-23T07:38:03.000000+0000	2025-07-23T08:10:40.000000+0000	56
2023-09-12T08:34:32.000000+0000	2023-09-12T08:53:31.000000+0000	39
2025-08-29T06:19:01.000000+0000	2025-08-29T06:37:03.000000+0000	37

Part D — Results

17. Exploratory analysis (development period)

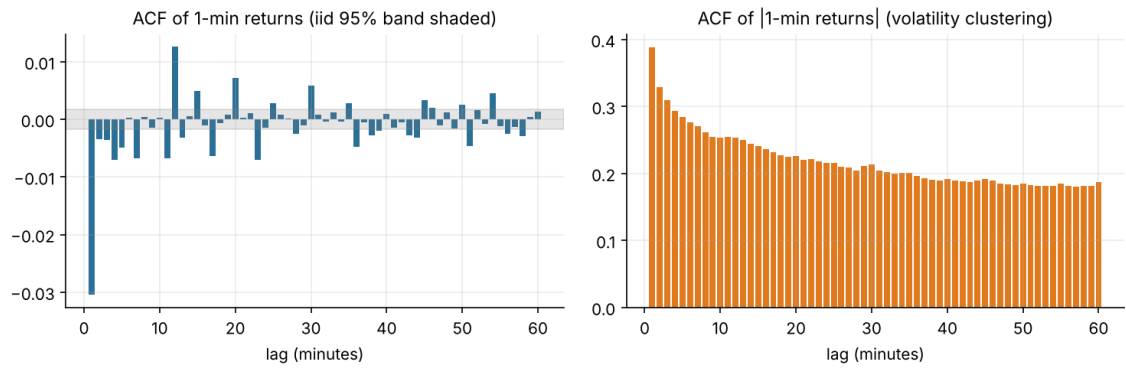


Price, daily realised volatility, volume and funding.



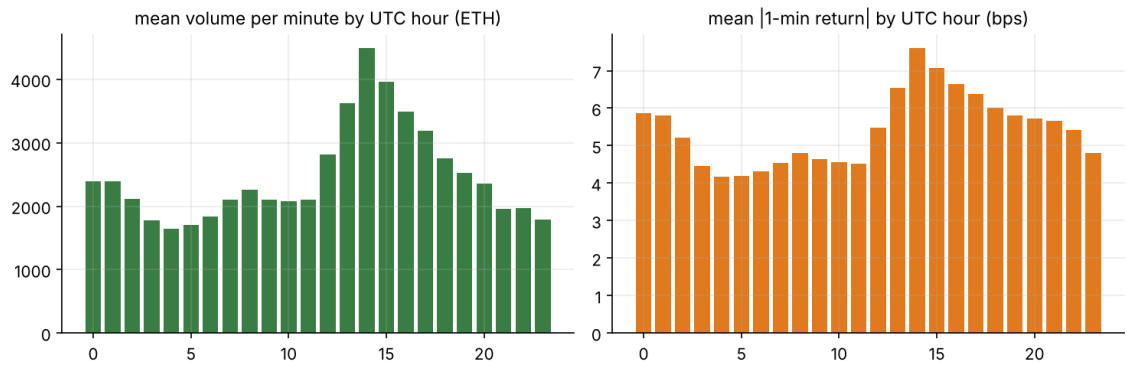
Standardised return distributions vs Normal (log density): extreme fat tails.

series	n	mean_bps	std_bps	skew	excess_kurtosis	p01_bps	p99_bps	share_zero
1m log return	1,360,739	0.007	8.924	-3.016	487.377	-23.636	23.684	0.012
15m forward return	1,360,740	0.158	32.883	-0.674	59.212	-92.044	91.861	0.001

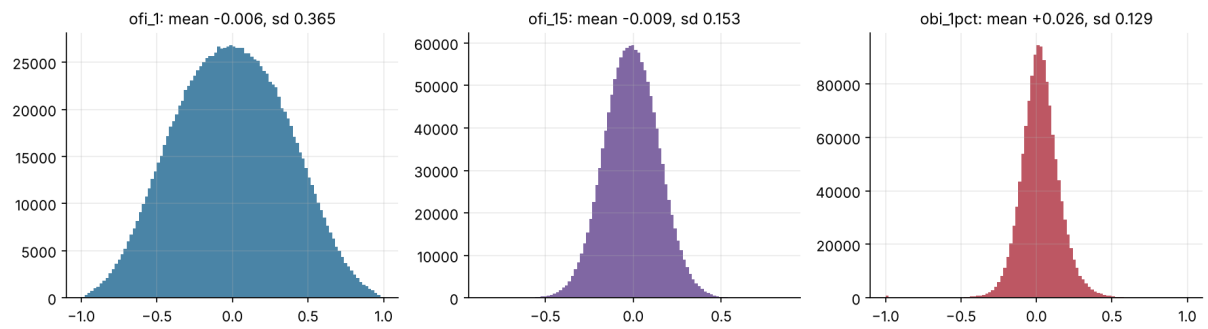


ACF of 1-min returns (lag-1 negative) and absolute returns (volatility clustering).

lag	ACF return	ACF return
1	-0.0305	0.3887
2	-0.0035	0.3294
3	-0.0036	0.3091
4	-0.0070	0.2935
5	-0.0049	0.2839
6	0.0003	0.2762
7	-0.0068	0.2708
8	0.0004	0.2609
9	-0.0015	0.2548
10	0.0003	0.2534

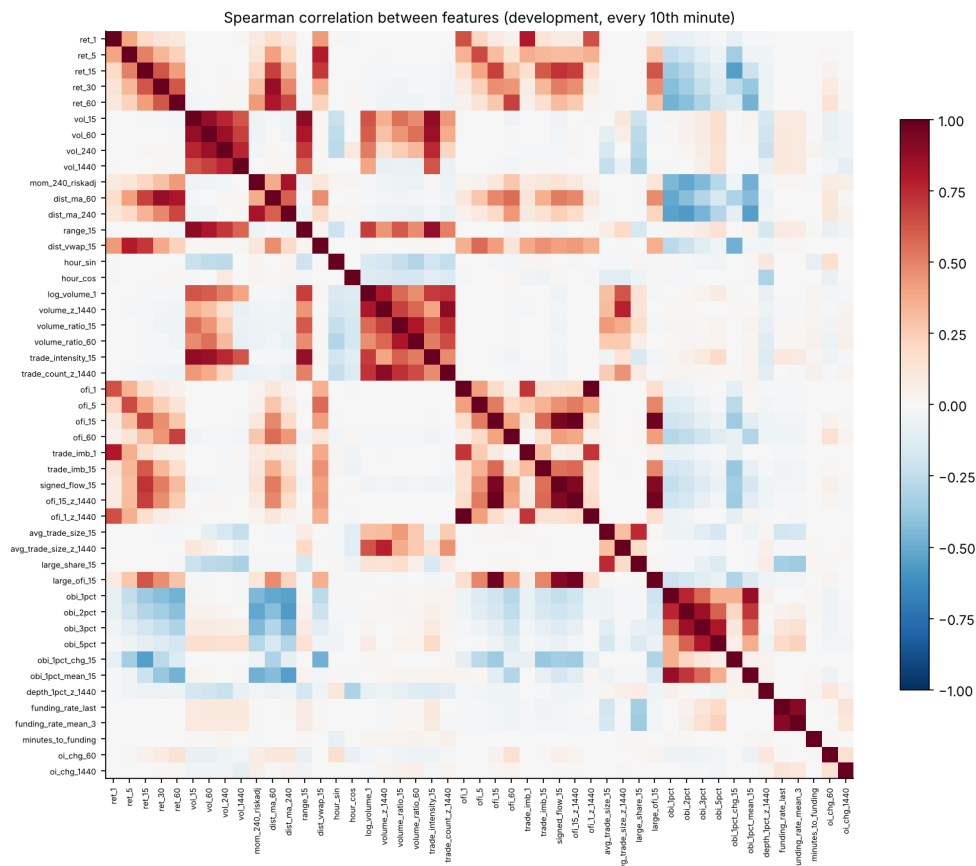


Intraday seasonality of volume and absolute returns (UTC).



Order-flow and book imbalance distributions.

feature	n	mean	sd	p05	p50	p95	acf1
ofi_1	1,360,687	-0.006	0.365	-0.603	-0.007	0.594	0.114
ofi_5	1,360,721	-0.008	0.226	-0.376	-0.009	0.361	0.822
ofi_15	1,360,716	-0.009	0.153	-0.258	-0.010	0.243	0.943
ofi_60	1,360,681	-0.009	0.086	-0.149	-0.010	0.132	0.986
trade_imb_15	1,360,716	2.8e-04	0.113	-0.176	4.2e-04	0.175	0.974
obi_1pct	1,306,853	0.026	0.129	-0.172	0.022	0.238	0.921
obi_2pct	1,306,853	0.052	0.122	-0.136	0.048	0.255	0.959
obi_3pct	1,306,853	0.064	0.114	-0.103	0.056	0.267	0.978
obi_5pct	1,306,853	0.096	0.113	-0.069	0.087	0.302	0.989



Spearman correlation among all 47 features.

18. Statistical alpha tests (development period)

Contemporaneous $\text{corr}(\text{OFI}_1, r_1) = 0.4668$; $\text{corr}(\text{OBI}_{1\%}, r_1) = -0.0734$; development rows 1,360,740.

Univariate predictive regressions (all 36 tests)

signal	h	n	Pearson	Spearman	β bps/sd	HAC t	Holm p	BH p	non-ovl t
ofi_1	5	1,360,687	-0.008	-0.016	-0.146	-9.739	7.4e-21	7.4e-21	-1.986
ofi_1	15	1,360,687	-0.005	-0.014	-0.170	-5.990	7.2e-08	2.5e-08	-1.094
ofi_1	30	1,360,687	-0.004	-0.012	-0.194	-4.765	6.2e-05	1.7e-05	-2.539
ofi_1	60	1,360,687	-0.004	-0.008	-0.237	-4.015	0.002	3.6e-04	-1.066
ofi_5	5	1,360,721	-0.009	-0.027	-0.170	-6.940	1.4e-10	7.1e-11	-3.631
ofi_5	15	1,360,721	-0.006	-0.025	-0.200	-3.819	0.004	6.9e-04	-0.908
ofi_5	30	1,360,721	-0.005	-0.023	-0.224	-2.897	0.105	0.015	0.115
ofi_5	60	1,360,721	-0.004	-0.019	-0.260	-2.336	0.468	0.054	0.327
ofi_15	5	1,360,716	-0.004	-0.031	-0.081	-2.820	0.130	0.017	-1.513
ofi_15	15	1,360,716	-0.004	-0.033	-0.148	-2.017	0.961	0.105	-3.538
ofi_15	30	1,360,716	-0.005	-0.034	-0.244	-2.055	0.917	0.103	-2.419
ofi_15	60	1,360,716	-0.003	-0.027	-0.203	-1.140	1.000	0.381	0.557
ofi_60	5	1,360,681	-0.002	-0.032	-0.035	-1.080	1.000	0.404	-0.997
ofi_60	15	1,360,681	-0.001	-0.037	-0.038	-0.420	1.000	0.837	-0.439
ofi_60	30	1,360,681	-0.001	-0.038	-0.061	-0.372	1.000	0.852	-0.748
ofi_60	60	1,360,681	-6.8e-04	-0.035	-0.045	-0.152	1.000	0.923	-0.644
trade_imb_15	5	1,360,716	-0.003	-0.022	-0.051	-2.499	0.324	0.041	-0.958
trade_imb_15	15	1,360,716	-0.003	-0.025	-0.086	-1.572	1.000	0.220	-2.668
trade_imb_15	30	1,360,716	-0.004	-0.027	-0.178	-1.894	1.000	0.121	-1.737
trade_imb_15	60	1,360,716	-0.003	-0.021	-0.214	-1.357	1.000	0.314	0.110
signed_flow_15	5	1,360,021	-9.3e-04	-0.035	-0.018	-0.227	1.000	0.913	-0.547
signed_flow_15	15	1,360,021	0.002	-0.038	0.076	0.456	1.000	0.834	0.853
signed_flow_15	30	1,360,021	9.8e-04	-0.038	0.045	0.205	1.000	0.913	1.378
signed_flow_15	60	1,360,021	7.6e-05	-0.030	0.005	0.016	1.000	0.988	-0.697
large_ofi_15	5	1,360,721	-0.003	-0.030	-0.057	-1.878	1.000	0.121	-0.849
large_ofi_15	15	1,360,721	-0.003	-0.031	-0.093	-1.197	1.000	0.378	0.684
large_ofi_15	30	1,360,721	-0.003	-0.033	-0.147	-1.162	1.000	0.381	-1.394
large_ofi_15	60	1,360,721	-3.8e-04	-0.025	-0.025	-0.129	1.000	0.923	0.700
obi_1pct	5	1,306,853	0.008	0.037	0.146	4.027	0.002	3.6e-04	3.288
obi_1pct	15	1,306,853	0.007	0.045	0.237	2.406	0.404	0.048	1.223
obi_1pct	30	1,306,853	0.004	0.045	0.180	0.979	1.000	0.454	0.313
obi_1pct	60	1,306,853	-0.001	0.037	-0.084	-0.250	1.000	0.913	-0.560
obi_5pct	5	1,306,853	0.006	0.019	0.121	2.946	0.093	0.014	2.806
obi_5pct	15	1,306,853	0.007	0.028	0.231	1.911	1.000	0.121	1.114
obi_5pct	30	1,306,853	0.006	0.032	0.294	1.244	1.000	0.366	0.751
obi_5pct	60	1,306,853	0.005	0.034	0.348	0.787	1.000	0.575	0.496

With controls (ret_15, ret_60, vol_60, volume_z_1440)

signal	h	n	β bps/sd	t	Holm p	t ret15	t ret60	t vol60	t volz
ofi_15	5	1,360,011	0.100	1.297	1.000	-2.232	-0.245	1.856	-0.507
ofi_15	15	1,360,011	-0.034	-0.250	1.000	-0.646	-0.457	2.224	-1.313
ofi_15	30	1,360,011	-0.172	-0.969	1.000	-0.122	-0.644	2.258	-1.924
ofi_15	60	1,360,011	0.044	0.176	1.000	-0.890	-0.410	2.417	-2.710
ofi_1	5	1,359,968	-0.112	-5.874	6.8e-08	-2.304	-0.391	1.854	-0.513
ofi_1	15	1,359,968	-0.147	-4.374	1.8e-04	-0.794	-0.477	2.228	-1.342
ofi_1	30	1,359,968	-0.178	-3.987	9.4e-04	-0.376	-0.633	2.262	-1.974
ofi_1	60	1,359,968	-0.198	-3.264	0.014	-0.936	-0.426	2.430	-2.818
obi_1pct	5	1,306,145	0.057	0.896	1.000	-2.139	-0.016	1.824	-0.417
obi_1pct	15	1,306,145	0.177	1.374	1.000	-0.676	-0.106	2.162	-1.272
obi_1pct	30	1,306,145	0.076	0.355	1.000	-0.411	-0.468	2.165	-1.994
obi_1pct	60	1,306,145	-0.326	-0.846	1.000	-0.973	-0.574	2.271	-2.694
trade_imb_15	5	1,360,011	0.064	1.351	1.000	-2.360	-0.249	1.855	-0.499
trade_imb_15	15	1,360,011	-0.005	-0.059	1.000	-0.791	-0.446	2.224	-1.318
trade_imb_15	30	1,360,011	-0.121	-0.984	1.000	-0.271	-0.643	2.256	-1.929
trade_imb_15	60	1,360,011	-0.068	-0.353	1.000	-0.846	-0.424	2.417	-2.708

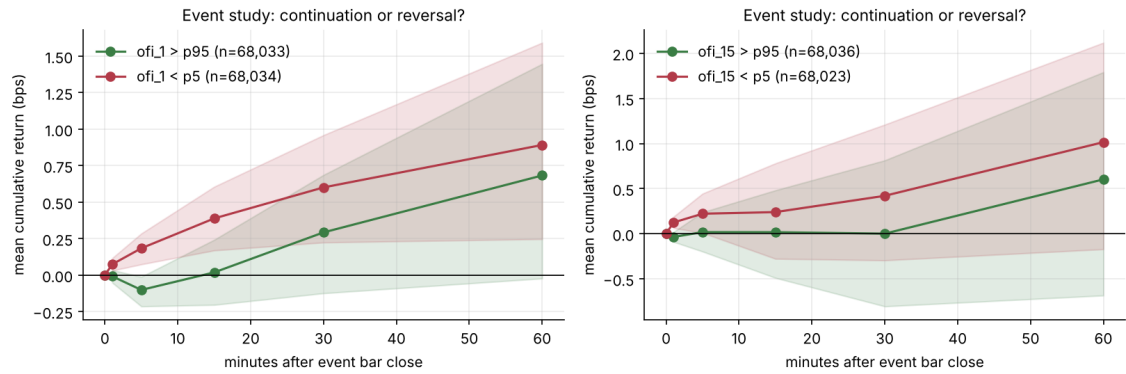
Decile top – bottom spreads (5-day block bootstrap)

signal	h	spread bps	SE	CI lo	CI hi	p	Holm p	monotonicity
ofi_15	5	-0.226	0.120	-0.456	0.016	0.060	0.418	-0.770
ofi_15	15	-0.371	0.316	-1.009	0.294	0.240	1.000	-0.891
ofi_15	30	-0.542	0.508	-1.500	0.339	0.287	1.000	-0.867
ofi_15	60	-0.446	0.639	-1.667	0.727	0.485	1.000	-0.697
ofi_1	5	-0.290	0.063	-0.413	-0.173	4.4e-06	5.3e-05	-0.879
ofi_1	15	-0.292	0.101	-0.469	-0.088	0.004	0.040	-0.867
ofi_1	30	-0.437	0.158	-0.710	-0.115	0.006	0.052	-0.830
ofi_1	60	-0.453	0.220	-0.898	-0.062	0.039	0.316	-0.794
obi_1pct	5	0.598	0.171	0.281	0.932	4.7e-04	0.005	0.624
obi_1pct	15	0.826	0.444	-0.031	1.688	0.063	0.418	0.709
obi_1pct	30	0.767	0.802	-0.872	2.307	0.339	1.000	-0.055
obi_1pct	60	-0.175	1.463	-3.050	2.581	0.905	1.000	-0.552



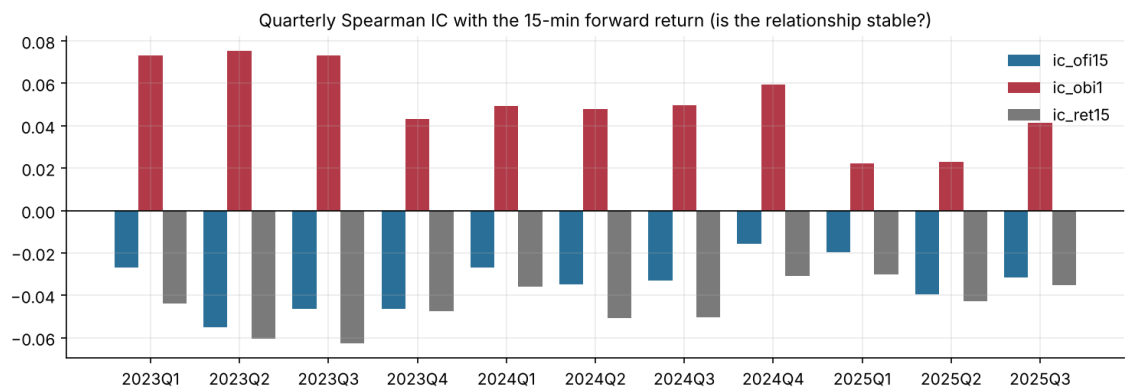
Event study

event	n	event bar	+1m	+5m	CI lo 5m	CI hi 5m	+15m	+60m	vol ratio 15m
ofi_1 > p95	68,033	5.398	-0.008	-0.101	-0.214	-0.011	0.019	0.683	0.664
ofi_1 < p5	68,034	-5.349	0.074	0.184	0.074	0.285	0.388	0.892	0.671
ofi_15 > p95	68,036	2.016	-0.036	0.018	-0.197	0.238	0.017	0.603	0.760
ofi_15 < p5	68,023	-2.141	0.125	0.222	0.010	0.445	0.240	1.015	0.801



Quarterly rank IC stability

quarter	IC ofi_15	IC obi_1%	IC ret_15	n
2023Q1	-0.0267	0.0732	-0.0437	44,640
2023Q2	-0.0550	0.0754	-0.0604	131,040
2023Q3	-0.0465	0.0730	-0.0627	132,480
2023Q4	-0.0463	0.0432	-0.0476	132,480
2024Q1	-0.0270	0.0493	-0.0359	131,040
2024Q2	-0.0348	0.0477	-0.0509	131,040
2024Q3	-0.0332	0.0496	-0.0503	132,480
2024Q4	-0.0158	0.0595	-0.0310	132,480
2025Q1	-0.0197	0.0224	-0.0302	129,600
2025Q2	-0.0395	0.0229	-0.0428	131,040
2025Q3	-0.0315	0.0415	-0.0351	132,420



19. Model results — all binary configurations (development walk-forward)

config	AUC mean	AUC sd	min fold	max fold	fold s > 0.5	log-loss gain	F1	ECE	IC
lgbm_A_price_h5	0.5303	0.0027	0.5257	0.5345	7	0.0014	0.5397	0.0075	0.0469
lgbm_C_plus_flow_h5	0.5307	0.0028	0.5255	0.5345	7	0.0013	0.5535	0.0092	0.0480
lgbm_E_all_h5	0.5308	0.0039	0.5252	0.5343	7	0.0013	0.5536	0.0104	0.0488
logit_A_price_h5	0.5270	0.0035	0.5214	0.5332	7	0.0008	0.5342	0.0074	0.0434
logit_C_plus_flow_h5	0.5271	0.0033	0.5215	0.5320	7	0.0009	0.5390	0.0087	0.0430
logit_E_all_h5	0.5275	0.0047	0.5211	0.5344	7	0.0009	0.5391	0.0096	0.0439
lgbm_A_price_h15	0.5375	0.0081	0.5281	0.5501	7	0.0019	0.5530	0.0115	0.0559
lgbm_B_price_volume_h15	0.5378	0.0077	0.5291	0.5506	7	0.0019	0.5569	0.0123	0.0566
lgbm_C_plus_flow_h15	0.5380	0.0071	0.5292	0.5493	7	0.0020	0.5521	0.0106	0.0564
lgbm_D_plus_book_h15	0.5391	0.0082	0.5296	0.5527	7	0.0020	0.5549	0.0132	0.0592
lgbm_E_all_h15	0.5388	0.0084	0.5284	0.5523	7	0.0020	0.5520	0.0145	0.0586
tune_lgbm_E_h15_g0	0.5388	0.0084	0.5284	0.5523	7	0.0020	0.5520	0.0145	0.0586
tune_lgbm_E_h15_g1	0.5371	0.0087	0.5279	0.5521	7	0.0010	0.5415	0.0207	0.0569
tune_lgbm_E_h15_g2	0.5385	0.0086	0.5277	0.5529	7	0.0019	0.5486	0.0148	0.0577
tune_lgbm_E_h15_g3	0.5362	0.0097	0.5264	0.5547	7	0.0010	0.5390	0.0211	0.0550
tune_lgbm_E_h15_g4	0.5359	0.0091	0.5269	0.5520	7	0.0009	0.5434	0.0215	0.0549
tune_lgbm_E_h15_g5	0.5322	0.0087	0.5242	0.5474	7	-0.0023	0.5303	0.0359	0.0504
tune_lgbm_E_h15_g6	0.5382	0.0084	0.5282	0.5517	7	0.0019	0.5476	0.0148	0.0574
tune_lgbm_E_h15_g7	0.5349	0.0090	0.5236	0.5503	7	0.0006	0.5366	0.0224	0.0534
logit_A_price_h15	0.5344	0.0076	0.5227	0.5426	7	0.0012	0.5492	0.0116	0.0514
logit_B_price_volume_h15	0.5346	0.0081	0.5220	0.5431	7	0.0012	0.5510	0.0108	0.0519
logit_C_plus_flow_h15	0.5349	0.0069	0.5231	0.5432	7	0.0013	0.5418	0.0119	0.0521
logit_D_plus_book_h15	0.5368	0.0083	0.5246	0.5479	7	0.0014	0.5406	0.0127	0.0557
logit_E_all_h15	0.5367	0.0081	0.5244	0.5476	7	0.0014	0.5395	0.0127	0.0554
rf_A_price_h15	0.5369	0.0079	0.5262	0.5480	7	0.0019	0.5428	0.0127	0.0549
rf_B_price_volume_h15	0.5371	0.0079	0.5269	0.5481	7	0.0019	0.5475	0.0125	0.0554
rf_C_plus_flow_h15	0.5368	0.0079	0.5264	0.5478	7	0.0019	0.5481	0.0127	0.0552
rf_D_plus_book_h15	0.5381	0.0083	0.5268	0.5486	7	0.0020	0.5482	0.0140	0.0573
rf_E_all_h15	0.5381	0.0087	0.5265	0.5496	7	0.0020	0.5481	0.0136	0.0572
lgbm_A_price_h30	0.5385	0.0111	0.5270	0.5578	7	0.0016	0.5639	0.0158	0.0583
lgbm_C_plus_flow_h30	0.5383	0.0108	0.5268	0.5571	7	0.0017	0.5671	0.0148	0.0588
lgbm_E_all_h30	0.5392	0.0116	0.5268	0.5598	7	0.0016	0.5690	0.0179	0.0605
logit_A_price_h30	0.5350	0.0092	0.5240	0.5460	7	0.0011	0.5707	0.0156	0.0522
logit_C_plus_flow_h30	0.5353	0.0072	0.5273	0.5465	7	0.0013	0.5654	0.0146	0.0533
logit_E_all_h30	0.5376	0.0091	0.5277	0.5526	7	0.0014	0.5617	0.0155	0.0573
lgbm_A_price_h60	0.5356	0.0118	0.5216	0.5493	7	0.0007	0.5726	0.0206	0.0534
lgbm_C_plus_flow_h60	0.5362	0.0128	0.5205	0.5508	7	0.0011	0.5725	0.0188	0.0548
lgbm_E_all_h60	0.5361	0.0133	0.5226	0.5535	7	0.0008	0.5717	0.0220	0.0541
logit_A_price_h60	0.5334	0.0127	0.5179	0.5468	7	0.0008	0.5773	0.0202	0.0498
logit_C_plus_flow_h60	0.5349	0.0109	0.5201	0.5468	7	0.0007	0.5824	0.0187	0.0520
logit_E_all_h60	0.5363	0.0108	0.5221	0.5493	7	0.0006	0.5711	0.0213	0.0548

LightGBM tuning grid

grid_id	num_leaves	min_child_samples	n_estimators	mean_log_loss	sd_log_loss	mean_auc	sd_auc	min_auc	mean_ll_improvement
0	15	500	200	0.6911	0.0012	0.5388	0.0084	0.5284	0.0020
2	15	5,000	200	0.6912	0.0013	0.5385	0.0086	0.5277	0.0019
6	63	5,000	200	0.6912	0.0013	0.5382	0.0084	0.5282	0.0019
1	15	500	600	0.6921	0.0014	0.5371	0.0087	0.5279	0.0010
3	15	5,000	600	0.6921	0.0017	0.5362	0.0097	0.5264	0.0010
4	63	500	200	0.6922	0.0014	0.5359	0.0091	0.5269	0.0009
7	63	5,000	600	0.6925	0.0017	0.5349	0.0090	0.5236	0.0006
5	63	500	600	0.6954	0.0023	0.5322	0.0087	0.5242	-0.0023

Primary-model selection (pre-declared: lowest mean validation log loss, set E, h = 15)

config	log loss	AUC	AUC sd
rf_E_all_h15	0.69108	0.53805	0.00870
lgbm_E_all_h15	0.69113	0.53878	0.00838
logit_E_all_h15	0.69167	0.53666	0.00814

Rule baselines (AUC per horizon over 7 folds)

rule	h	mean AUC	min	max
flow_ofi1	5	0.4920	0.4887	0.4970
flow_rule_ofi15z	5	0.4836	0.4793	0.4871
momentum_ret15	5	0.4793	0.4746	0.4831
prior_only	5	0.5000	0.5000	0.5000
flow_ofi1	15	0.4925	0.4869	0.4988
flow_rule_ofi15z	15	0.4812	0.4700	0.4911
momentum_ret15	15	0.4745	0.4625	0.4857
prior_only	15	0.5000	0.5000	0.5000
flow_ofi1	30	0.4923	0.4874	0.4970
flow_rule_ofi15z	30	0.4834	0.4718	0.4926
momentum_ret15	30	0.4761	0.4630	0.4824
prior_only	30	0.5000	0.5000	0.5000
flow_ofi1	60	0.4949	0.4914	0.5008
flow_rule_ofi15z	60	0.4878	0.4856	0.4928
momentum_ret15	60	0.4802	0.4764	0.4874
prior_only	60	0.5000	0.5000	0.5000

Alternative targets (per fold)

fold	macro AUC	UP vs DOWN AUC	log loss	prior	share neutral	IC
F1	0.6108	0.5499	1.0608	1.1714	0.3581	0.0650
F2	0.6431	0.5485	1.0227	1.0956	0.4245	0.0653
F3	0.6246	0.5490	1.0455	1.1142	0.3895	0.0642
F4	0.6007	0.5312	1.0669	1.1517	0.3190	0.0382
F5	0.6048	0.5423	1.0693	1.1431	0.3126	0.0566
F6	0.5957	0.5349	1.0702	1.1179	0.3460	0.0470
F7	0.6156	0.5632	1.0332	1.0701	0.4496	0.0870
fold	MAE	MAE (0)	RMSE	RMSE (mean)	IC	sign hit
F1	24.4930	24.4305	37.9076	37.8321	0.0219	0.5061

fold	MAE	MAE (0)	RMSE	RMSE (mean)	IC	sign hit
F2	21.0330	20.9955	34.9184	34.8715	0.0060	0.5030
F3	21.7772	21.7313	32.0296	31.9606	0.0019	0.4978
F4	27.1890	27.1633	42.2852	42.1433	0.0250	0.5081
F5	27.4577	27.4494	42.1660	42.1474	0.0252	0.5077
F6	23.6659	23.6516	34.6981	34.7116	0.0105	0.4982
F7	17.5178	17.5017	25.6075	25.5817	0.0027	0.5007

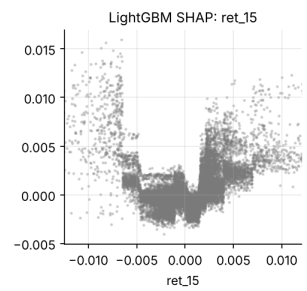
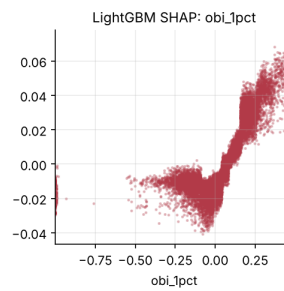
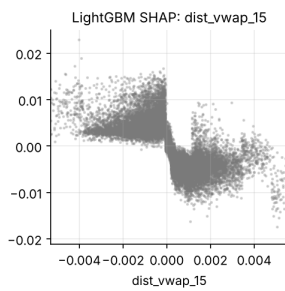
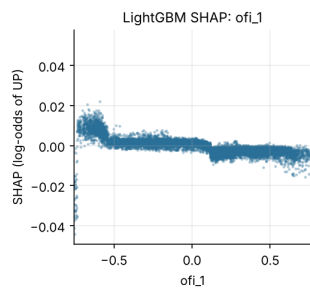
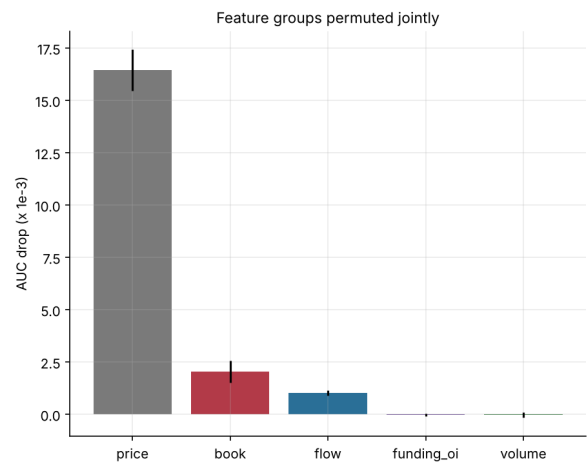
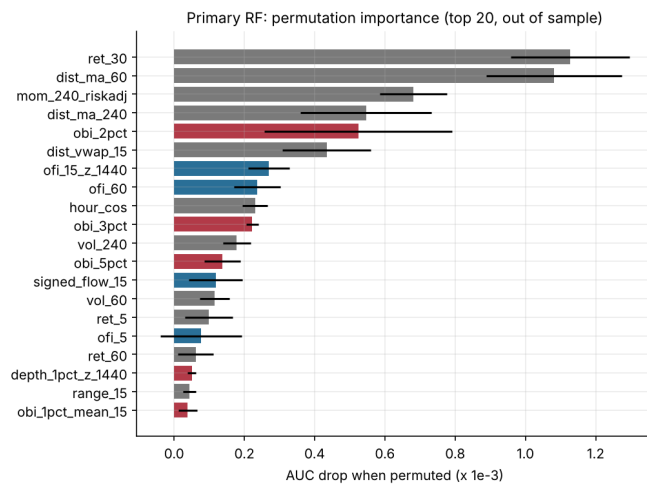
Per-fold results of the primary model

fold	valid start	train rows	valid rows	AUC	log loss	prior	acc	F1	ECE	IC
F1	2024-03-01	35,039	132,465	0.5386	0.6909	0.6931	0.5247	0.5534	0.0120	0.0550
F2	2024-06-01	43,871	132,465	0.5472	0.6899	0.6931	0.5337	0.5567	0.0084	0.0721
F3	2024-09-01	52,703	131,025	0.5412	0.6905	0.6931	0.5294	0.5494	0.0083	0.0638
F4	2024-12-01	61,439	129,585	0.5265	0.6930	0.6931	0.5168	0.5365	0.0192	0.0360
F5	2025-03-01	70,079	132,465	0.5342	0.6912	0.6930	0.5223	0.5348	0.0151	0.0478
F6	2025-06-01	78,911	132,465	0.5292	0.6922	0.6929	0.5215	0.5369	0.0204	0.0441
F7	2025-09-01	87,743	43,185	0.5496	0.6898	0.6931	0.5366	0.5692	0.0119	0.0814

20. Interpretability (out of sample, fold F6: train to 2025-05-30, explain 2025-06 → 2025-08)

Primary RF on this fold: AUC 0.5292 (LightGBM 0.5312). Permutation = AUC drop when a feature (or a whole group, jointly) is shuffled, 3 repeats on 200,000 validation rows. SHAP = exact TreeSHAP of the runner-up LightGBM.

group	n	AUC drop	sd	log-loss increase		
price	16	0.01644	0.00099	0.00144		
book	7	0.00204	0.00053	-0.00002		
flow	13	0.00103	0.00011	-0.00005		
funding_oi	5	-0.00003	0.00005	-4.8e-06		
volume	6	-0.00004	0.00012	-0.00002		
group	Σ mean SHAP			gain share		
price	0.1854			0.5017		
book	0.0756			0.1865		
flow	0.0472			0.1252		
volume	0.0273			0.0822		
funding_oi	0.0237			0.1043		
feature	group	perm AUC drop	sd	RF impurity	mean SHAP	LGBM gain share
ret_30	price	0.00113	0.00017	0.05591	0.00688	0.01443
dist_ma_60	price	0.00108	0.00019	0.06827	0.04351	0.10788
mom_240_riskadj	price	0.00068	0.00010	0.03542	0.01238	0.03566
dist_ma_240	price	0.00055	0.00019	0.05653	0.04696	0.12513
obi_2pct	book	0.00052	0.00027	0.04435	0.02019	0.04542
dist_vwap_15	price	0.00043	0.00013	0.05313	0.00439	0.00906
ofi_15_z_1440	flow	0.00027	0.00006	0.01854	0.00095	0.00328
ofi_60	flow	0.00024	0.00007	0.02475	0.00889	0.02414
hour_cos	price	0.00023	0.00004	0.01273	0.01332	0.04030
obi_3pct	book	0.00022	0.00002	0.02455	0.01313	0.02964
vol_240	price	0.00018	0.00004	0.01518	0.01229	0.03561
obi_5pct	book	0.00014	0.00005	0.01338	0.00366	0.01755
signed_flow_15	flow	0.00012	0.00008	0.02113	0.00742	0.01601
vol_60	price	0.00012	0.00004	0.01109	0.00147	0.00988
ret_5	price	0.00010	0.00007	0.03147	0.00570	0.00848
ofi_5	flow	0.00008	0.00012	0.01984	0.00498	0.00865
ret_60	price	0.00006	0.00005	0.04194	0.00943	0.02007
depth_1pct_z_1440	book	0.00005	0.00001	0.01120	0.00440	0.02024
range_15	price	0.00004	0.00002	0.01361	0.00484	0.00827
obi_1pct_mean_15	book	0.00004	0.00003	0.02562	0.00822	0.02325
funding_rate_mean_3	funding_oi	0.00003	0.00002	0.00960	0.00163	0.01914
avg_trade_size_15	flow	0.00003	6.3e-06	0.01367	0.00644	0.01475
funding_rate_last	funding_oi	0.00003	0.00001	0.00899	0.00575	0.02113
avg_trade_size_z_1440	flow	0.00002	0.00002	0.00939	0.00145	0.00808
ofi_15	flow	0.00001	0.00002	0.01511	0.00368	0.00642



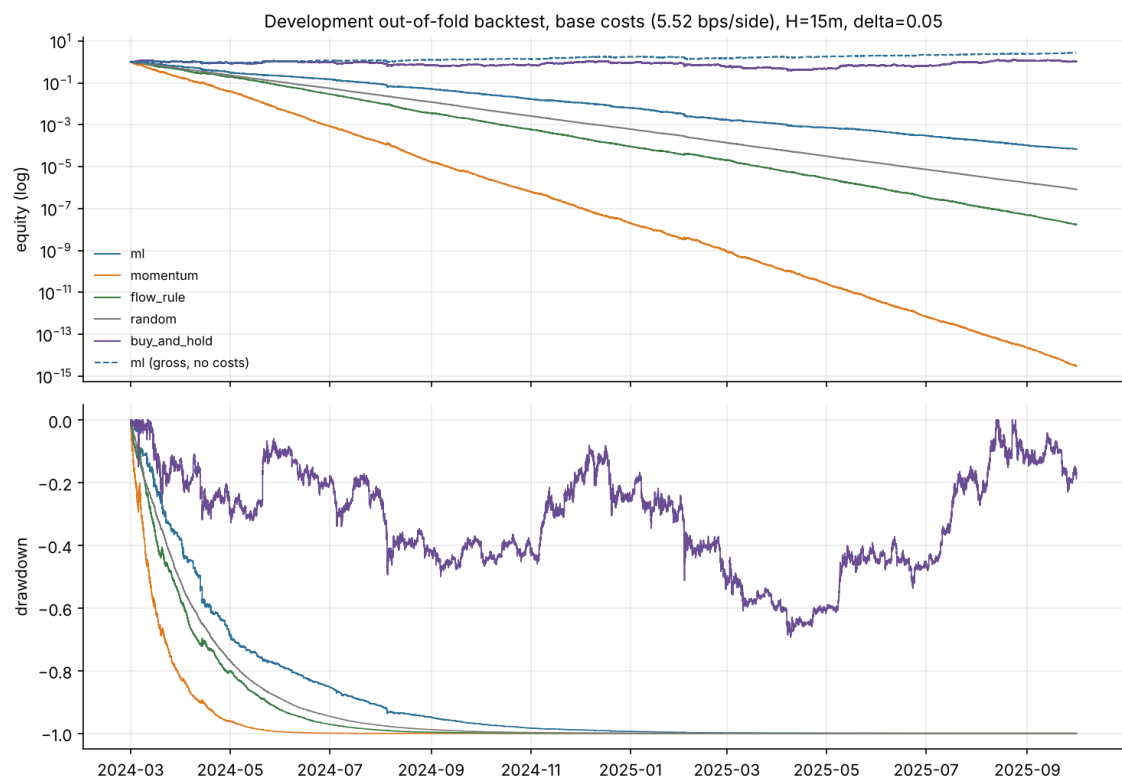
Standardised logistic coefficients (top 15)

feature	group	coefficient
ofi_15	flow	-0.1792
large_ofi_15	flow	0.0902
ofi_15_z_1440	flow	0.0594
mom_240_riskadj	price	-0.0586
log_volume_1	volume	0.0492
trade_count_z_1440	volume	-0.0425
volume_ratio_15	volume	0.0405
dist_ma_60	price	-0.0364
large_share_15	flow	-0.0341
ofi_60	flow	-0.0338
obi_2pct	book	0.0329
trade_intensity_15	volume	-0.0327
obi_1pct_mean_15	book	0.0312
signed_flow_15	flow	0.0278
obi_5pct	book	-0.0258

21. Development backtest and threshold selection (out-of-fold, 2024-03 → 2025-09)

δ	active	Sharpe base	return base	Sharpe gross	Σ gross	BE bps	trades	turnover/day
0.005	0.910	-24.515	-1.000	1.274	1.111	0.296	758,470	63.784
0.010	0.821	-24.270	-1.000	1.289	1.107	0.301	684,381	62.599
0.020	0.650	-23.588	-1.000	1.295	1.037	0.306	542,051	57.867
0.030	0.497	-22.118	-1.000	1.528	1.097	0.370	414,528	50.694
0.050	0.261	-17.819	-1.000	2.092	1.107	0.573	217,566	33.086

strategy	cost	total return	Sharpe	max DD	trades	win rate	avg trade bps	break-even bps	turnover/day
ml	gross	173.2%	2.09	-29.0%	217,566	56.0%	0.76	0.57	33.09
momentum	gross	-79.8%	-1.78	-83.1%	833,147	47.8%	-0.25	-0.25	99.50
flow_rule	gross	-59.2%	-1.91	-66.1%	278,576	47.4%	-0.45	-0.27	53.07
random	gross	-10.2%	-0.66	-17.2%	216,795	49.9%	-0.07	-0.04	43.46
buy_and_hold	gross	4.6%	0.32	-69.4%	1	-	-	-	1.7e-03
ml	low	-97.8%	-7.10	-97.9%	217,566	48.5%	-4.28	0.57	33.09
momentum	low	-100.0%	-19.14	-100.0%	833,147	39.3%	-5.30	-0.25	99.50
flow_rule	low	-100.0%	-18.80	-100.0%	278,576	38.5%	-5.50	-0.27	53.07
random	low	-99.8%	-41.76	-99.8%	216,795	41.2%	-5.12	-0.04	43.46
buy_and_hold	low	4.6%	0.32	-69.4%	1	-	-	-	1.7e-03
ml	base	-100.0%	-17.82	-100.0%	217,566	39.4%	-10.28	0.57	33.09
momentum	base	-100.0%	-37.96	-100.0%	833,147	30.1%	-11.30	-0.25	99.50
flow_rule	base	-100.0%	-37.01	-100.0%	278,576	29.2%	-11.50	-0.27	53.07
random	base	-100.0%	-88.71	-100.0%	216,795	31.7%	-11.12	-0.04	43.46
buy_and_hold	base	4.5%	0.32	-69.4%	1	-	-	-	1.7e-03
ml	high	-100.0%	-23.00	-100.0%	217,566	35.1%	-13.28	0.57	33.09
momentum	high	-100.0%	-46.68	-100.0%	833,147	26.2%	-14.30	-0.25	99.50
flow_rule	high	-100.0%	-45.27	-100.0%	278,576	25.2%	-14.50	-0.27	53.07
random	high	-100.0%	-111.00	-100.0%	216,795	27.6%	-14.12	-0.04	43.46
buy_and_hold	high	4.5%	0.32	-69.4%	1	-	-	-	1.7e-03



Ablation economics (development out-of-fold, 6 frozen)

config	tradable AUC	AUC	Sharpe gross	return gross	Sharpe base	return base	BE bps	trades	avg gross bps
lgbm_A_price_h15	0.540	0.538	1.660	115.9%	-13.676	-99.9%	0.588	212,400	0.623
lgbm_B_price_volume_h15	0.540	0.538	2.057	168.7%	-13.980	-100.0%	0.701	221,529	0.747
lgbm_C_plus_flow_h15	0.540	0.538	2.244	184.5%	-14.046	-99.9%	0.748	206,458	0.836
lgbm_D_plus_book_h15	0.540	0.539	2.872	298.8%	-14.683	-100.0%	0.898	239,824	0.937
lgbm_E_all_h15	0.540	0.539	2.889	332.4%	-14.093	-100.0%	0.932	248,781	0.952
logit_A_price_h15	0.537	0.534	1.147	64.7%	-8.253	-99.0%	0.639	127,301	0.712
logit_B_price_volume_h15	0.537	0.535	1.541	99.7%	-8.231	-98.9%	0.830	127,551	0.933
logit_C_plus_flow_h15	0.537	0.535	1.661	113.0%	-11.911	-99.8%	0.651	168,514	0.766
logit_D_plus_book_h15	0.539	0.537	2.331	202.7%	-13.699	-99.9%	0.785	213,522	0.853
logit_E_all_h15	0.538	0.537	2.236	193.2%	-13.575	-99.9%	0.763	213,618	0.825
rf_A_price_h15	0.540	0.537	1.523	106.8%	-18.133	-100.0%	0.423	224,486	0.558
rf_B_price_volume_h15	0.541	0.537	1.622	112.9%	-18.147	-100.0%	0.450	215,005	0.601
rf_C_plus_flow_h15	0.540	0.537	1.852	140.6%	-17.447	-100.0%	0.522	200,975	0.731
rf_D_plus_book_h15	0.541	0.538	1.939	158.4%	-17.642	-100.0%	0.542	219,328	0.722
rf_E_all_h15	0.541	0.538	2.092	173.2%	-17.819	-100.0%	0.573	217,566	0.765

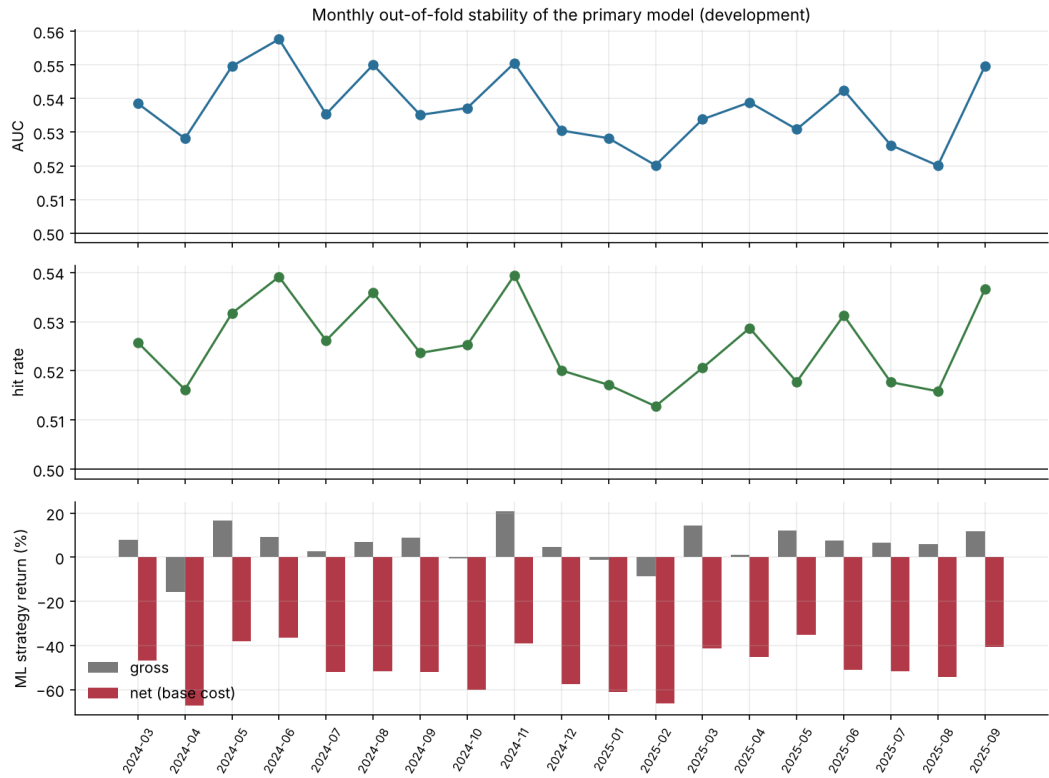
Edge by confidence (EXPLORATORY, non-binding)

min P-0.5	signals	share	mean gross bps	hit rate	round trip bps
0.000	833,654	1.000	0.192	0.528	11.046
0.020	542,051	0.650	0.287	0.540	11.046
0.050	217,566	0.261	0.765	0.560	11.046
0.075	69,527	0.083	1.850	0.579	11.046
0.100	9,566	0.011	2.965	0.590	11.046
0.125	46	5.5e-05	-38.611	0.565	11.046
0.150	0	0.000	-	-	11.046
0.200	0	0.000	-	-	11.046

Monthly stability and volatility regimes (development)

month	n	AUC	log loss	prior	hit rate	Σ net	Σ gross
2024-03	44,640	0.5385	0.6909	0.6929	0.5258	-0.4690	0.0797
2024-04	43,200	0.5281	0.6925	0.6931	0.5162	-0.6723	-0.1566
2024-05	44,625	0.5496	0.6893	0.6931	0.5317	-0.3823	0.1655
2024-06	43,200	0.5576	0.6886	0.6931	0.5391	-0.3636	0.0913
2024-07	44,640	0.5354	0.6915	0.6931	0.5262	-0.5204	0.0253
2024-08	44,625	0.5500	0.6896	0.6931	0.5359	-0.5164	0.0692
2024-09	43,200	0.5351	0.6913	0.6931	0.5236	-0.5193	0.0899
2024-10	44,640	0.5372	0.6911	0.6931	0.5252	-0.6019	-0.0040
2024-11	43,185	0.5505	0.6891	0.6928	0.5394	-0.3906	0.2073
2024-12	44,640	0.5305	0.6923	0.6930	0.5201	-0.5761	0.0452
2025-01	44,640	0.5282	0.6927	0.6931	0.5172	-0.6109	-0.0136
2025-02	40,305	0.5202	0.6942	0.6931	0.5128	-0.6619	-0.0861
2025-03	44,640	0.5338	0.6913	0.6931	0.5206	-0.4141	0.1449
2025-04	43,200	0.5389	0.6908	0.6931	0.5287	-0.4525	0.0123
2025-05	44,625	0.5309	0.6915	0.6922	0.5178	-0.3532	0.1202
2025-06	43,200	0.5425	0.6904	0.6931	0.5313	-0.5103	0.0754
2025-07	44,640	0.5262	0.6926	0.6921	0.5177	-0.5169	0.0645
2025-08	44,625	0.5201	0.6935	0.6929	0.5159	-0.5444	0.0587
2025-09	43,185	0.5496	0.6898	0.6931	0.5366	-0.4062	0.1181

regime	cost	minutes	AUC	Σ net	net bps/min	turnover
1_low	base	52,111	0.5460	-0.6120	-0.1174	1,063
2_mid	base	165,330	0.5347	-2.1496	-0.1300	3,750
3_high	base	616,319	0.5361	-6.7208	-0.1090	14,345
1_low	gross	52,111	0.5460	-0.0250	-0.0048	1,063
2_mid	gross	165,330	0.5347	-0.0787	-0.0048	3,750
3_high	gross	616,319	0.5361	1.2017	0.0195	14,345



22. Final holdout results (2025-10-01 → 2026-09-30, frozen, evaluated once)

metric (pooled)						value					
n						525,540					
base_rate						0.4980					
auc						0.5436					
log_loss						0.6902					
log_loss_baseline						0.6931					
brier						0.2485					
accuracy						0.5314					
precision						0.5290					
recall						0.5380					
f1						0.5334					
ece						0.0028					
pred_up_share						0.5064					
ic_spearman						0.0624					
tradable AUC						0.5480					
holdout rows						525,600					

block	start	train rows	test rows	AUC	log loss	prior	acc	F1	ECE	IC
F1	2025-10-01	90,623	132,465	0.5363	0.6912	0.6931	0.5271	0.5025	0.0127	0.0473
F2	2026-01-01	99,455	129,585	0.5488	0.6896	0.6931	0.5365	0.5541	0.0068	0.0708
F3	2026-04-01	108,095	131,025	0.5491	0.6896	0.6930	0.5345	0.5430	0.0090	0.0726
F4	2026-07-01	116,831	132,465	0.5412	0.6904	0.6931	0.5276	0.5322	0.0083	0.0607

Calibration (equal-count deciles)

bin	mean predicted	observed up share	n
0	0.4302	0.4316	52,554
1	0.4566	0.4546	52,554
2	0.4723	0.4677	52,554
3	0.4846	0.4832	52,554
4	0.4954	0.4961	52,554
5	0.5058	0.5056	52,554
6	0.5164	0.5179	52,554
7	0.5281	0.5236	52,554
8	0.5427	0.5339	52,554
9	0.5688	0.5653	52,554

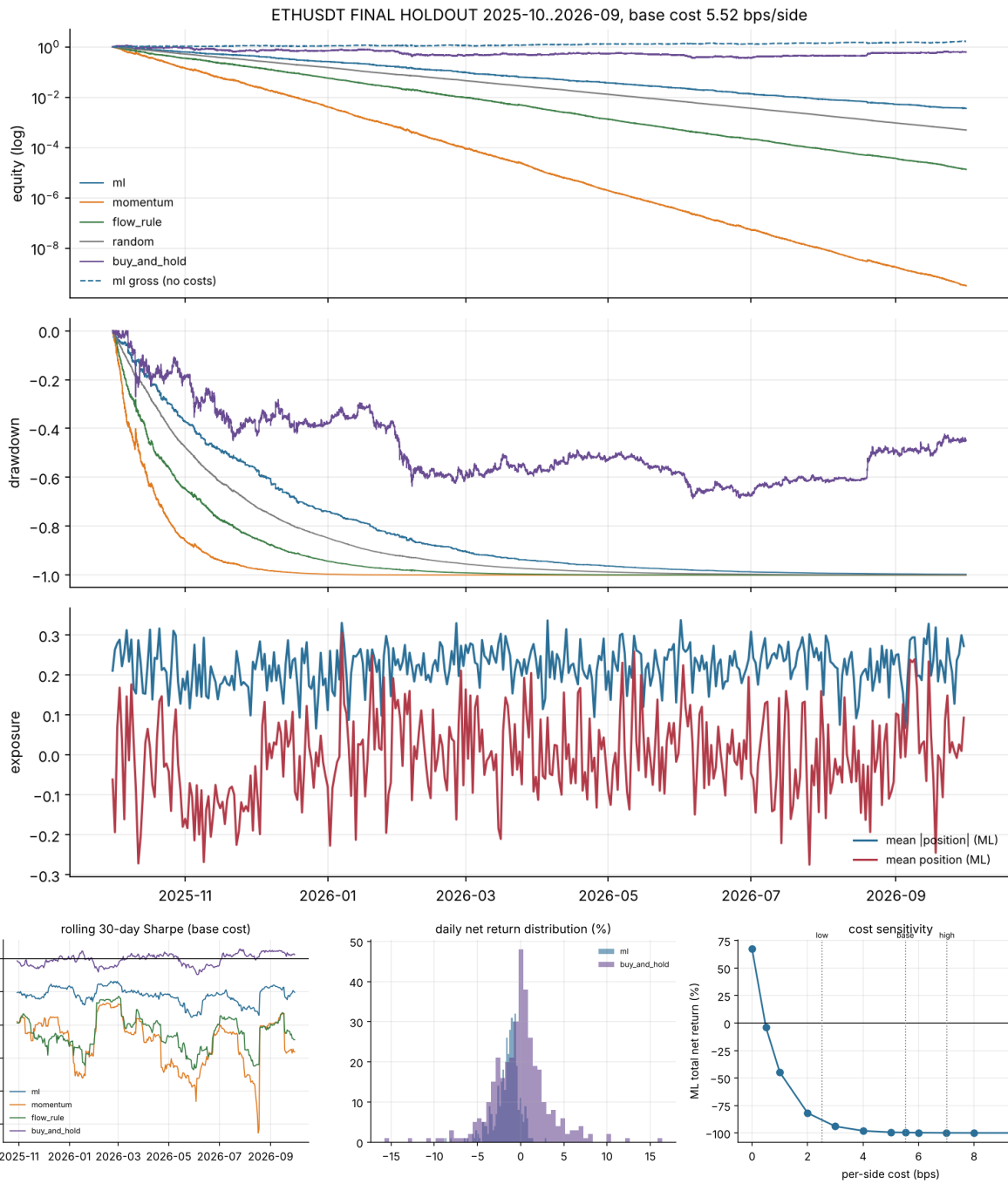
Backtest — all strategies and cost scenarios

(a) return and risk

strategy	cost	total return	annual return	annual vol	Sharpe	Sortino	max DD	Calmar
ml	gross	67.5%	67.5%	27.0%	2.03	2.73	-12.8%	5.26
momentum	gross	-78.1%	-78.1%	45.0%	-3.18	-4.29	-81.4%	-0.96
flow_rule	gross	-51.9%	-51.9%	25.1%	-2.81	-3.80	-53.2%	-0.98
random	gross	7.2%	7.2%	7.9%	0.91	1.43	-6.4%	1.12
buy_and_hold	gross	-36.9%	-36.9%	63.1%	-0.48	-0.67	-68.7%	-0.54
ml	low	-89.8%	-89.8%	27.0%	-8.35	-8.16	-90.3%	-0.99
momentum	low	-100.0%	-100.0%	46.9%	-22.84	-15.13	-100.0%	-1.00
flow_rule	low	-99.6%	-99.6%	25.7%	-21.33	-14.73	-99.6%	-1.00
random	low	-96.8%	-96.8%	7.9%	-43.37	-17.53	-96.8%	-1.00
buy_and_hold	low	-36.9%	-36.9%	63.1%	-0.48	-0.67	-68.7%	-0.54
ml	base	-99.6%	-99.6%	27.4%	-20.37	-14.03	-99.6%	-1.00
momentum	base	-100.0%	-100.0%	49.3%	-44.11	-17.59	-100.0%	-1.00
flow_rule	base	-100.0%	-100.0%	27.0%	-41.43	-17.43	-100.0%	-1.00
random	base	-99.9%	-99.9%	8.1%	-94.16	-18.72	-99.9%	-1.00
buy_and_hold	base	-36.9%	-36.9%	63.1%	-0.48	-0.67	-68.7%	-0.54
ml	high	-99.9%	-99.9%	27.8%	-26.11	-15.45	-99.9%	-1.00
momentum	high	-100.0%	-100.0%	50.6%	-53.94	-18.03	-100.0%	-1.00
flow_rule	high	-100.0%	-100.0%	27.8%	-50.47	-17.90	-100.0%	-1.00
random	high	-100.0%	-100.0%	8.2%	-118.15	-18.86	-100.0%	-1.00
buy_and_hold	high	-36.9%	-36.9%	63.1%	-0.48	-0.67	-68.7%	-0.54

(b) trading activity

strategy	cost	trades	win rate	profit factor	avg trade bps	break-ev en bps	turnover r/day	exposure	funding
ml	gross	117,461	57.1%	1.057	0.695	0.493	30.408	0.222	0.1%
momentum	gross	525,022	47.2%	0.961	-0.410	-0.389	100.844	0.610	0.1%
flow_rule	gross	175,617	46.3%	0.941	-0.600	-0.371	51.938	0.312	0.1%
random	gross	117,194	49.9%	1.009	0.092	0.052	38.023	0.094	0.0%
buy_and_hold	gross	1	-	-	-	-	0.003	1.000	-2.5%
ml	low	117,461	47.3%	0.700	-4.351	0.493	30.408	0.222	0.1%
momentum	low	525,022	36.3%	0.598	-5.456	-0.389	100.844	0.610	0.1%
flow_rule	low	175,617	35.2%	0.573	-5.646	-0.371	51.938	0.312	0.1%
random	low	117,194	39.0%	0.625	-4.954	0.052	38.023	0.094	0.0%
buy_and_hold	low	1	-	-	-	-	0.003	1.000	-2.5%
ml	base	117,461	36.0%	0.426	-10.351	0.493	30.408	0.222	0.1%
momentum	base	525,022	26.1%	0.353	-11.456	-0.389	100.844	0.610	0.1%
flow_rule	base	175,617	25.0%	0.332	-11.646	-0.371	51.938	0.312	0.1%
random	base	117,194	28.1%	0.363	-10.954	0.052	38.023	0.094	0.0%
buy_and_hold	base	1	-	-	-	-	0.003	1.000	-2.5%
ml	high	117,461	30.9%	0.333	-13.351	0.493	30.408	0.222	0.1%
momentum	high	525,022	22.2%	0.277	-14.456	-0.389	100.844	0.610	0.1%
flow_rule	high	175,617	21.0%	0.259	-14.646	-0.371	51.938	0.312	0.1%
random	high	117,194	23.8%	0.282	-13.954	0.052	38.023	0.094	0.0%
buy_and_hold	high	1	-	-	-	-	0.003	1.000	-2.5%



cost bps/side	total return	Sharpe
0.000	67.5%	2.028
0.500	-3.9%	-0.030
1.000	-44.8%	-2.090
2.000	-81.8%	-6.204
3.000	-94.0%	-10.294
4.000	-98.0%	-14.339
5.000	-99.3%	-18.321
5.523	-99.6%	-20.372
6.000	-99.8%	-22.222
7.000	-99.9%	-26.026
8.000	-100.0%	-29.720
10.000	-100.0%	-36.733

Monthly stability (holdout)

month	AUC	log loss	prior	hit rate	Σ net	Σ gross
2025-10	0.5338	0.6916	0.6931	0.5276	-0.4599	0.0641
2025-11	0.5214	0.6929	0.6930	0.5144	-0.3595	0.0695
2025-12	0.5517	0.6892	0.6931	0.5389	-0.5151	-0.0215
2026-01	0.5510	0.6893	0.6931	0.5405	-0.5066	0.0104
2026-02	0.5464	0.6901	0.6931	0.5343	-0.4598	0.0485
2026-03	0.5492	0.6896	0.6931	0.5345	-0.5199	0.0611
2026-04	0.5589	0.6880	0.6931	0.5403	-0.4369	0.0596
2026-05	0.5416	0.6906	0.6930	0.5300	-0.4792	0.0290
2026-06	0.5480	0.6902	0.6927	0.5334	-0.5241	-0.0056
2026-07	0.5372	0.6911	0.6931	0.5258	-0.4960	0.0679
2026-08	0.5465	0.6895	0.6931	0.5316	-0.4577	0.0333
2026-09	0.5411	0.6905	0.6931	0.5255	-0.3685	0.1296

Weekly net returns of the ML strategy (base cost)

week	net	week	net
2025-09-29	-5.15%	2026-04-06	-11.91%
2025-10-06	-11.85%	2026-04-13	-11.97%
2025-10-13	-8.74%	2026-04-20	-4.31%
2025-10-20	-12.50%	2026-04-27	-12.67%
2025-10-27	-10.06%	2026-05-04	-11.90%
2025-11-03	-13.15%	2026-05-11	-12.37%
2025-11-10	-5.87%	2026-05-18	-10.10%
2025-11-17	-7.40%	2026-05-25	-9.17%
2025-11-24	-7.23%	2026-06-01	-15.01%
2025-12-01	-16.46%	2026-06-08	-5.97%
2025-12-08	-10.06%	2026-06-15	-14.12%
2025-12-15	-13.21%	2026-06-22	-14.26%
2025-12-22	-8.71%	2026-06-29	-13.17%
2025-12-29	-7.08%	2026-07-06	-8.72%
2026-01-05	-9.65%	2026-07-13	-9.96%
2026-01-12	-10.94%	2026-07-20	-13.19%
2026-01-19	-15.30%	2026-07-27	-11.44%
2026-01-26	-10.13%	2026-08-03	-6.19%
2026-02-02	-15.43%	2026-08-10	-8.43%
2026-02-09	-10.78%	2026-08-17	-15.51%
2026-02-16	-6.33%	2026-08-24	-11.09%
2026-02-23	-18.49%	2026-08-31	-4.73%
2026-03-02	-12.29%	2026-09-07	-15.11%
2026-03-09	-13.02%	2026-09-14	-10.68%
2026-03-16	-9.06%	2026-09-21	-3.47%
2026-03-23	-10.09%	2026-09-28	-3.58%
2026-03-30	-10.31%		

23. Robustness and replication

Holdout feature ablation (frozen procedure)

model	set	AUC	tradable AUC	log loss	IC	Sharpe gross	return gross	Sharpe net	return net	BE bps
logit	A_price	0.535	0.538	0.692	0.050	0.248	3.6%	-7.061	-86.5%	0.180
lgbm	A_price	0.541	0.545	0.691	0.058	1.861	55.5%	-15.279	-98.2%	0.591
rf	A_price	0.543	0.547	0.690	0.061	1.405	44.1%	-19.770	-99.7%	0.367
logit	B_price_volume	0.534	0.537	0.692	0.049	0.278	4.4%	-7.125	-86.7%	0.199
lgbm	B_price_volume	0.541	0.545	0.691	0.057	1.578	44.9%	-15.708	-98.3%	0.499
rf	B_price_volume	0.542	0.547	0.690	0.060	1.386	40.4%	-20.576	-99.6%	0.348
logit	C_plus_flow	0.536	0.541	0.691	0.051	0.893	22.9%	-11.714	-96.2%	0.378
lgbm	C_plus_flow	0.541	0.545	0.691	0.057	2.108	62.9%	-16.562	-98.5%	0.614
rf	C_plus_flow	0.543	0.547	0.690	0.060	1.408	42.0%	-19.945	-99.6%	0.359
logit	D_plus_book	0.534	0.537	0.694	0.050	2.171	102.0%	-8.941	-96.2%	1.047
lgbm	D_plus_book	0.541	0.546	0.691	0.058	1.978	57.6%	-18.184	-99.0%	0.530
rf	D_plus_book	0.543	0.548	0.690	0.062	1.810	57.9%	-20.690	-99.7%	0.440
logit	E_all	0.534	0.537	0.694	0.050	2.200	104.7%	-8.868	-96.1%	1.066
lgbm	E_all	0.542	0.546	0.690	0.059	2.103	61.9%	-18.790	-99.0%	0.548
rf	E_all	0.544	0.548	0.690	0.062	2.028	67.5%	-20.372	-99.6%	0.493

Holdout horizons

model	set	h	AUC	log loss	prior	IC
logit	A_price	5	0.5337	0.6918	0.6931	0.0532
lgbm	A_price	5	0.5372	0.6910	0.6931	0.0569
logit	C_plus_flow	5	0.5344	0.6914	0.6931	0.0527
lgbm	C_plus_flow	5	0.5367	0.6911	0.6931	0.0564
logit	E_all	5	0.5306	0.6926	0.6931	0.0485
lgbm	E_all	5	0.5371	0.6910	0.6931	0.0574
logit	A_price	30	0.5316	0.6920	0.6931	0.0459
lgbm	A_price	30	0.5399	0.6908	0.6931	0.0566
logit	C_plus_flow	30	0.5328	0.6917	0.6931	0.0452
lgbm	C_plus_flow	30	0.5388	0.6910	0.6931	0.0549
logit	E_all	30	0.5303	0.6948	0.6931	0.0434
lgbm	E_all	30	0.5400	0.6908	0.6931	0.0554
logit	A_price	60	0.5294	0.6923	0.6931	0.0412
lgbm	A_price	60	0.5372	0.6914	0.6931	0.0530
logit	C_plus_flow	60	0.5304	0.6921	0.6931	0.0397
lgbm	C_plus_flow	60	0.5357	0.6916	0.6931	0.0493
logit	E_all	60	0.5255	0.6954	0.6931	0.0351
lgbm	E_all	60	0.5370	0.6913	0.6931	0.0539

Holdout volatility regimes and confirmatory test

regime	cost	minutes	AUC	Σ net	turnover	
1_low	base	76,923	0.5499	-0.9225	1,458	
2_mid	base	126,901	0.5475	-1.3564	2,759	
3_high	base	321,776	0.5411	-3.3043	6,881	
1_low	gross	76,923	0.5499	-0.1171	1,458	
2_mid	gross	126,901	0.5475	0.1676	2,759	
3_high	gross	321,776	0.5411	0.4963	6,881	
signal	h	n	β bps/sd	HAC t	p	Holm p
ofi_1	5	525,595	-0.0690	-2.7360	0.0062	0.0373
ofi 1	15	525,585	-0.0551	-1.1936	0.2326	0.2326

signal	h	n	β bps/sd	HAC t	p	Holm p
obi_1pct	5	521,401	0.1348	2.0274	0.0426	0.1705
obi_1pct	15	521,391	0.3116	1.5826	0.1135	0.2270
ofi_15	5	525,595	-0.1005	-2.1306	0.0331	0.1656
ofi_15	15	525,585	-0.2289	-1.8841	0.0596	0.1787

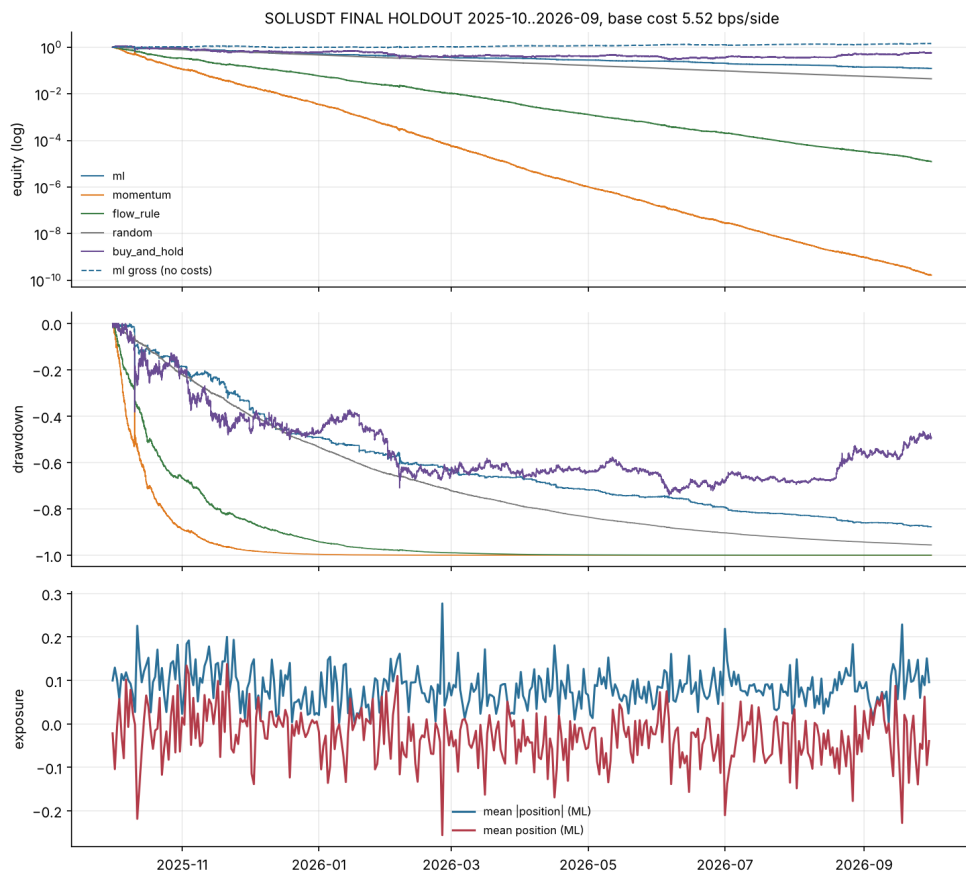
SOLUSDT replication (frozen settings)

Development walk-forward:

set	AUC	sd	IC	tradable AUC	Sharpe gross	return gross	Sharpe base	return base	BE bps
A_price	0.527	0.006	0.039	0.529	4.009	393.7%	-8.542	-97.8%	1.732
C_plus_flow	0.527	0.005	0.041	0.530	4.484	371.6%	-8.091	-95.6%	1.934
E_all	0.529	0.006	0.043	0.530	4.498	427.2%	-7.248	-94.9%	2.080
signal	h	n	β bps/sd		HAC t		p		
ofi_1	5	1,360,747	-0.2504		-11.3739		5.6e-30		
ofi_1	15	1,360,747	-0.3751		-9.4137		4.8e-21		
obi_1pct	5	1,304,038	0.3200		6.1058		1.0e-09		
obi_1pct	15	1,304,038	0.5511		4.1067		4.0e-05		
ofi_15	5	1,360,781	-0.2274		-5.3913		7.0e-08		
ofi_15	15	1,360,781	-0.4557		-4.2202		2.4e-05		

Holdout: AUC 0.5343, tradable AUC 0.5373, ECE 0.0079.

block	start	n	AUC	LL	prior	IC
F1	2025-10-01	132,465	0.5249	0.6923	0.6931	0.0328
F2	2026-01-01	129,585	0.5396	0.6907	0.6928	0.0567
F3	2026-04-01	131,025	0.5391	0.6908	0.6928	0.0597
F4	2026-07-01	132,465	0.5328	0.6911	0.6928	0.0509
strategy	cost	return	Sharpe	MDD	trades	BE bps
ml	gross	43.6%	1.74	-25.6%	44,412	0.89
momentum	gross	-90.3%	-4.36	-91.7%	515,695	-0.61
flow_rule	gross	-63.8%	-3.07	-64.6%	175,221	-0.52
random	gross	2.6%	0.52	-3.9%	44,634	0.05
buy_and_hold	gross	-42.6%	-0.51	-74.1%	1	-3,430
ml	base	-87.7%	-8.78	-87.8%	44,412	0.89
momentum	base	-100.0%	-40.59	-100.0%	515,695	-0.61
flow_rule	base	-100.0%	-33.29	-100.0%	175,221	-0.52
random	base	-95.6%	-57.83	-95.6%	44,634	0.05
buy_and_hold	base	-42.6%	-0.51	-74.1%	1	-3,430
signal	h	n	β bps/sd	HAC t	p	Holm p
ofi_1	5	525,595	-0.0591	-2.0158	0.0438	0.1753
ofi_1	15	525,585	-0.0848	-1.5780	0.1146	0.3437
obi_1pct	5	519,967	-0.0022	-0.0170	0.9865	1.0000
obi_1pct	15	519,957	-0.0488	-0.1493	0.8813	1.0000
ofi_15	5	525,595	-0.1674	-3.0225	0.0025	0.0150
ofi_15	15	525,585	-0.3688	-2.6963	0.0070	0.0351



24. Significance and data-snooping accounting

quantity	value	detail
Holdout AUC	0.5436	block-bootstrap SE 0.0022, 95% CI [0.5394, 0.5480], $z = 19.6$
Bonferroni	43 configurations	z needed 3.05 → passes: True
Gross Sharpe	2.03	95% CI [0.32, 3.75] over 365 days
Net Sharpe	-20.37	95% CI [-23.05, -18.27]
Deflated-Sharpe hurdle	3.28	expected max Sharpe of 215 zero-skill variants over 1.0 years: the gross Sharpe does not clear it
Experiments	43 model configs	41 binary classifiers, 5 thresholds, 36 univariate tests

25. Conclusions

Question	Answer (evidence)
1. Is there order-flow predictability?	Yes, statistically; no, economically. 1-min OFI predicts a reversal with Holm-adjusted p down to $7e-21$ in development and $p = 0.037$ on the holdout (5 min), but the effect is ≈ 0.07 - 0.24 bps per standard deviation.
2. Does order flow add information beyond price and volume?	Marginally in development, not robustly on the holdout. It adds $+0.0001$ to $+0.002$ AUC in development; holdout increments are within ± 0.002 and negative for logistic regression with book features.
3. Does ML improve the signal?	Yes, as a forecaster. ML is far better than the rule baselines (rules AUC < 0.5 ; ML 0.54). Trees beat logistic regression by ≈ 0.002 - 0.010 AUC on the holdout. It does not create an economically viable signal.
4. Does the signal survive out of sample?	Yes. Holdout AUC is 0.5436 [0.5394, 0.5480], 12/12 months > 0.5 , calibration ECE 0.003, tradable AUC 0.548.
5. Does it survive realistic costs?	No. The break-even cost is 0.49 bps per side against 5.52 bps base (2.52 bps even at the low scenario). Net -99.6% .
6a. Is it stable across time?	Yes for prediction. Monthly AUC ranges from 0.520 to 0.558 in development and 0.521 to 0.559 on the holdout, with no decay. Development quarterly IC for OFI-15 is negative in all 11 quarters.
6b. Is it stable across volatility regimes?	Prediction yes. AUC is 0.541-0.550 on the holdout across regimes. Profit is negative net of costs in every regime.
6c. Does it replicate on another asset?	Yes, qualitatively. SOL holdout AUC is 0.534, with a stronger OFI reversal in development ($t = -11.4$). It is also not tradable (break-even 0.89 bps per side).
Strongest limitation	Taker-only cost model and band-level book data. A maker or sub-second strategy cannot be evaluated with these data.

Bottom line. There is strong, stable, out-of-sample statistical evidence that one-minute ETH perpetual price paths mean-revert, and that taker order flow is mildly informative about that reversal. The predictable component is worth about 0.5 bps per side per trade, roughly one tenth of the cost of taking liquidity. **The signal is statistically significant and economically untradable at realistic costs.** Higher accuracy does not mean higher profit.

Part E — Appendices

A. Frozen methodology (docs/final_methodology.md, verbatim)

Frozen at **2026-10-07T14:21:08+00:00**, before any holdout row was evaluated.

scripts/freeze_methodology.py generated this file from config/config.yaml and the development-period selection outputs. reports/tables/frozen_parameters.json is the machine-readable copy that scripts/run_holdout.py reads. Every holdout access is logged in reports/holdout_access_log.jsonl together with the SHA-256 of this file. If the file changes after the holdout is opened, the hashes in the log will no longer match.

Environment: Python 3.13.16, packages pinned in requirements.txt. Seeds: model seed 20231 (ethof.models.SEED), bootstrap seeds fixed in code.

1. Data and periods

- Instrument: Binance USDS-M **ETHUSDT perpetual** (public archive, SHA-256-verified files).
- Full period 2023-03-01 → 2026-09-30 UTC. **Development** 2023-03-01 → 2025-09-30. **Holdout** 2025-10-01 → 2026-10-01 (exclusive).
- Research frequency: complete 1-minute grid. Bar t covers $[t_s, t_s+1 \text{ min})$.
- Data treatment: as in docs/data_quality_report.md. Nothing is deleted for being extreme. Book features are missing when the snapshot is older than 120 s, where age is measured from the last snapshot whose values *changed* (this covers the frozen feed). Open interest ≤ 0 is set to missing, and $\pm\text{inf}$ ratios are set to missing.

2. Timeline

Step	Time
features of row t	data stamped $< \tau_t = t_s + 1 \text{ min}$ (trades of bars $\leq t$; last book snapshot $< \tau_t$; settled funding $\leq \tau_t$; OI created $\leq \tau_t - 5 \text{ min}$)
signal	at τ_t
execution	during bar $t+1$ at its VWAP (taker); never at the signal price
holding	15 bars per signal (staggered: net position = mean of the last 15 signals)
exit	VWAP of bar $t+1+15$ (implicitly, through the staggered position)
target	$(P[t+15] - P[t]) / P[t]$, P = last trade price at or before the bar close

3. Features (47 in the primary set E)

Group	n	Features
price	16	ret_1, ret_5, ret_15, ret_30, ret_60, vol_15, vol_60, vol_240, vol_1440, mom_240_riskadj, dist_ma_60, dist_ma_240, range_15, dist_vwap_15, hour_sin, hour_cos
volume	6	log_volume_1, volume_z_1440, volume_ratio_15, volume_ratio_60, trade_intensity_15, trade_count_z_1440
flow	13	ofi_1, ofi_5, ofi_15, ofi_60, trade_imb_1, trade_imb_15, signed_flow_15, ofi_15_z_1440, ofi_1_z_1440, avg_trade_size_15, avg_trade_size_z_1440, large_share_15, large_ofi_15
book	7	obi_1pct, obi_2pct, obi_3pct, obi_5pct, obi_1pct_chg_15, obi_1pct_mean_15, depth_1pct_z_1440
funding_oi	5	funding_rate_last, funding_rate_mean_3, minutes_to_funding, oi_chg_60, oi_chg_1440

The order-book features are **percentage-band** imbalances (cumulative resting quantity within $\pm 1/2/3/5 \%$ of the price). They are **not** top-N level features, and no best bid/ask or spread information is used. Trailing windows only. Large-trade threshold is ≥ 10 ETH (rule: the size bucket closest to the median per-minute p95 trade size in 2023-03..2024-02).

Ablation sets: A = price; B = A + volume; C = B + flow; D = C + book; E = D + funding/OI.

4. Target

- Primary: binary $\text{up}_{15} = 1\{\text{fwd_ret}_{15} > 0\}$. Secondary (reported, not used for trading): 3-class with neutral band ± 11 bps (= base round-trip cost); regression on fwd_ret_{15} ; horizons 5/30/60.

5. Validation and preprocessing

- Expanding walk-forward, 3-month blocks, first validation block 2024-03-01, purge of training rows whose label ends after (block start – 1 day embargo), training stride 5 minutes.
- Imputation and scaling happen inside sklearn pipelines fitted per training fold. LightGBM uses native missing-value handling.

6. Model (frozen)

- Selection rule (pre-declared): the lowest mean validation log loss on set E, $h = 15$, among logit / RF / LightGBM.
- **Selected:** `rf_E_all_h15` (model class `rf`).
- LightGBM hyper-parameters (8-point grid, chosen by mean validation log loss): `{"num_leaves": 15, "min_child_samples": 500, "n_estimators": 200}`; other parameters are fixed in `ethof.models.LGBM_DEFAULT`. Logit $C = 0.1$. RF: `{"n_estimators": 200, "min_samples_leaf": 500, "max_depth": 10}`, stride 15.
- Holdout procedure: four 3-month blocks. Before each block the frozen model is re-fitted on all data from 2023-03-01 to (block start – 1 day) (the same expanding scheme as development). No parameter, feature or threshold is revisited.

7. Trading rule (frozen)

- LONG if $P(\text{up}) > 0.5 + \delta$, SHORT if $P(\text{up}) < 0.5 - \delta$, FLAT otherwise. $\delta = 0.05$, chosen as the highest out-of-fold net Sharpe at base cost among $[0.005, 0.01, 0.02, 0.03, 0.05]$ (development out-of-fold, 2024-03 → 2025-09).
- Position sizing: each signal is a sub-position of 1/15 of capital held 15 bars; net $|\text{position}| \leq 1$; no leverage; shorting allowed (perpetual).
- Benchmarks: buy-and-hold (long 1×), momentum `sign(ret_15)`, order-flow rule (long if `ofi_15_z_1440 > 1.0`, short if `< -1.0`), random signals with the ML strategy's activity rate (seeded).

8. Costs, slippage, funding (per side, on traded notional)

Scenario	fee (bps)	slippage (bps)	half spread (bps)	total per side (bps)
low	2.0	0.5	0.023	2.523
base	4.5	1.0	0.023	5.523
high	5.0	2.0	0.023	7.023

Sources: Binance USD-M VIP0 fees are 2 bps maker and 5 bps taker, with a 10% discount when paid in BNB. The half spread was measured from `bookTicker` on 3 development days (`reports/tables/spread_measurement.csv`). Slippage is an assumption, bracketed by the low and high scenarios. **Base** is the headline scenario. Funding: the realised rate at each settlement is charged to the position held over the settlement (longs pay a positive rate, shorts receive it). The break-even cost is $(\text{gross P\&L} + \text{funding}) / \text{total turnover}$, expressed per side.

9. Metrics and pre-declared secondary holdout evaluations

- Classification: AUC, log loss vs prior, Brier, accuracy, precision/recall/F1, ECE, Spearman IC, and *tradable AUC* (against the VWAP-to-VWAP return actually capturable).
- Trading: total and annualised return, volatility, Sharpe and Sortino (daily, 365-day year), max drawdown, Calmar, win rate, profit factor, average trade (bps, net of a full round trip), turnover, exposure, funding, break-even cost.
- Also evaluated on the holdout, with the same frozen procedure: ablation sets $A-E \times \{\text{logit, LightGBM, RF}\}$ at $h = 15$; horizons 5/30/60 (sets A, C, E $\times$ logit and LightGBM); a confirmatory HAC test of 1-minute OFI reversal; volatility regimes (tercile cut-offs $4.838\text{e-}04$ / $6.694\text{e-}04$ on `vol_1440`, estimated on 2023-03..2024-02); monthly stability; SOLUSDt replication with identical settings.

B. Stage 1 dataset selection (verbatim)

Project: **Ethereum Order-Flow Alpha: Short-Horizon Return Prediction and Cost-Aware Trading Strategy**

Everything below was checked against the live archive on 2026-10-07 by downloading sample files and running HTTP HEAD requests for each day. Nothing here is a model result.

1. Candidates considered

Source	Trades w/ aggressor side	Historical order book	Accessible here	Verdict
Coinbase Exchange ETH-USD (REST /products/ETH-USD/trades)	Yes (side)	No historical L2 (live websocket only)	No (blocked by the sandbox's egress policy, HTTP 403)	Rejected: pulling years of data 1,000 trades per request is not practical, and it has no historical book
Tardis.dev (Coinbase/Binance L2 + trades)	Yes	Yes (true L2 snapshots)	No (403). It's also a paid service; only the first day of each month is free	Rejected: not reproducible for free
Binance Spot ETHUSDT (data.binance.vision)	Yes (is_buyer_maker)	No order-book files for spot	Yes	Secondary: robustness only
Binance USDS-M Futures ETHUSDT perpetual (data.binance.vision)	Yes (is_buyer_maker)	Yes, but coarse: bookDepth (see §3)	Yes	Recommended

2. Recommended dataset

- **Exchange / venue:** Binance USDS-M Futures
- **Instrument:** ETHUSDT perpetual swap (linear, USDT-margined)
- **Source:** Binance Public Data archive, <https://data.binance.vision> (documentation: <https://github.com/binance/binance-public-data>)
- **Proposed period: 2023-03-01 → 2026-09-30 (43 months)**, all timestamps in UTC
- **Primary research frequency:** 1-minute bars built from trades (about 1.88M bars); 5-minute bars as a robustness check
- **Access:** free, no API key. Plain HTTPS downloads of daily and monthly ZIP files. Each file has a .CHECKSUM (SHA-256) sidecar.

Files used

Dataset	URL pattern	Fields (verified)	Cadence	Size
aggTrades	/data/futures/um/monthly/aggTrades/ETHUSDT/ETHUSDT-aggTrades-YYYY-MM.zip	agg_trade_id, price, quantity, first_trade_id, last_trade_id, transact_time (ms), is_buyer_maker	tick (~1M agg trades/day)	350–730 MB zipped per month, about 20 GB in total
bookDepth	/data/futures/um/daily/bookDepth/ETHUSDT/ETHUSDT-bookDepth-YYYY-MM-DD.zip	timestamp, percentage ($\pm 1.. \pm 5$), depth (ETH), notional (USD)	snapshot about every 30 s (2,881 per day)	~0.5 MB per day, ~0.6 GB in total
fundingRate	/data/futures/um/monthly/fundingRate/ETHUSDT/...	funding time, rate	8 h	negligible
metrics (optional)	/data/futures/um/daily/metrics/ETHUSDT/...	open interest, long/short ratios, taker buy/sell volume ratio	5 min	negligible
bookTicker (validation only)	/data/futures/um/daily/bookTicker/ETHUSDT/...	best bid/ask price and qty, tick-by-tick	tick	3.4 GB CSV per day; the archive ends 2024-03-30

Replication asset (same schema): **SOLUSDT** perpetual. I confirmed that bookDepth exists for it.

3. What the data do and do not contain (honest limitations)

1. **Trade direction is explicit, not inferred.** `is_buyer_maker = true` means the aggressor was a seller. Buy/sell volume therefore needs no Lee-Ready or tick-rule classification. One caveat: `aggTrades` merges fills from the same taker order at the same price, so "number of trades" counts aggregated trades. Large-trade statistics should be read with that in mind.

2. **The order book is not a level-by-level L2 book.** `bookDepth` gives *cumulative* resting quantity within $\pm 1\%$, $\pm 2\%$, ..., $\pm 5\%$ of the price, snapshotted about every 30 seconds. So:
 - OBI can be computed at **depth bands of 1%, 2%, 3%, 4% and 5%**, but **not** at "top-1 / top-5 / top-10 levels". $\pm 1\%$ of ETH is about 300 ticks, far deeper than the touch.
 - ****Best bid, best ask, mid and spread are *not* in `bookDepth`.** The only historical top-of-book file (`bookTicker`) stopped on 2024-03-30, and each day is about 3.4 GB.
 - Plan: (a) use `bookDepth` band OBI as the order-book feature set; (b) measure the actual quoted spread from `bookTicker` on a sample of days (Q1 2024) so that the cost model uses an *empirical* spread rather than an assumption; (c) report this mismatch in the README rather than labelling band depth as "top-N levels".
 - Snapshots are every 30 s, and a 1-minute bar's book feature will use **the last snapshot at or before the bar close**. That is an as-of backward join, so no forward fill crosses the bar boundary.
3. **Depth coverage gaps** (from HEAD checks of all 1,374 days from 2023-01-01 to 2026-10-05):
 - Jan-Mar 2023 has missing or truncated days. For example, 2023-01-15 has only 7 snapshots. The first unbroken 30-day run starts on 2023-02-10, **which is why the study starts on 2023-03-01**.
 - After that, only one short day was found (in 2024-06). It will be flagged, not imputed.
 - 2025-04/05 files are smaller on disk, but they hold complete days (2,881 snapshots). They just compress better. **Correction (Stage 2/3):** this was wrong. The files are small because the archive *repeats one snapshot* from 2025-04-16 10:31 to 2025-05-19 10:26 (94,730 identical snapshots). The feed is frozen, not complete. The pipeline now detects unchanged snapshots and treats those minutes as stale. See `docs/data_quality_report.md`.
4. **Timestamp unit changes.** Spot archives switch from milliseconds to **microseconds on 2025-01-01**. The futures files I checked are in ms. The parser will detect the unit per file and check it.
5. **This is a perpetual, not spot.** Returns include basis moves, and holding positions incurs **funding** every 8 h. Funding is downloaded and charged in the backtest. Shorting is native, which suits a long/short signal. Binance futures is the highest-volume ETH venue, so it is a defensible venue for price discovery. It is still *one* venue, though, and there is no cross-venue consolidated flow.
6. **Licensing and access.** The data are published by Binance for free public download and come under Binance's terms of use. Raw files **will not be committed** to the repo. The repo will ship the download script, a manifest of files and SHA-256 checksums, so anyone can rebuild the data.
7. **Survivorship and selection.** ETH was chosen in advance, not because of results. The period was chosen only because of data coverage, before looking at any returns.

4. Practical size plan

- About 20 GB of zipped futures trades against about 26 GB of free disk in this container. The pipeline will stream **one month at a time**: download → verify SHA-256 → parse → aggregate to 1-minute bars → write Parquet → (optionally) delete the raw zip. No rows are dropped. Only the storage of the raw zip is temporary, and the manifest records every file processed.
- Processed 1-minute bars are about 1.9M rows. That fits in memory easily.

5. Pre-registered split (fixed now, before any returns are examined)

Segment	Period	Use
Development (walk-forward)	2023-03-01 → 2025-09-30 (31 months)	EDA, features, expanding walk-forward CV with purge/embargo, model and threshold selection
Final holdout	2025-10-01 → 2026-09-30 (12 months)	Touched once, after the methodology is frozen

Note: EDA and the statistical alpha tests (Stage 4) will also be run on the development period only.

C. Data schema (verbatim)

Raw inputs (Binance public archive, USD-M futures, ETHUSD perpetual)

<https://data.binance.vision/data/futures/um/{daily|monthly}/{dataset}/ETHUSD/ETHUSD-{dataset}-{period}.zip> plus a .zip.CHECKSUM sidecar (<sha256> <filename>). Each download is verified before use and logged in data/manifest.csv.

Dataset	Granularity used	Columns	Notes
aggTrades	daily (YYYY-MM-DD)	agg_trade_id, price, quantity, first_trade_id, last_trade_id, transact_time, is_buyer_maker	transact_time in epoch ms for futures (the unit is detected per file; spot switched to μ s in 2025). is_buyer_maker=true $\Rightarrow$ the seller was the taker. One aggTrade = the fills of one taker order at one price.
bookDepth	daily	timestamp, percentage, depth, notional	Snapshot about every 30 s. percentage $\in \{-5\ldots-1, 1\ldots5\}$: cumulative quantity (depth, ETH) and value (notional, USD) resting within k % below (bids, negative) / above (asks, positive) the reference price. Not level-by-level L2. No best bid/ask or spread. Timestamps are YYYY-MM-DD HH:MM:SS, which we read as UTC and check in the Stage 2 report.
fundingRate	monthly	calc_time, funding_interval_hours, last_funding_rate	calc_time ms.
metrics	daily	create_time, symbol, sum_open_interest, sum_open_interest_value, count_toptrader_long_short_ratio, sum_toptrader_long_short_ratio, count_long_short_ratio, sum_taker_long_short_vol_ratio	5-minute. Downloaded; whether it is used is decided in Stage 3.

Processed outputs (data/processed/, git-ignored)

bars_1m/ETHUSD-bars_1m-YYYY-MM.parquet

One row per UTC minute. The grid is complete: minutes without trades are kept, never dropped. A bar labelled ts covers [ts, ts+1min).

Column(s)	Definition
ts	bar start, Datetime[us, UTC]
open, high, low, close	first / max / min / last trade price in the bar; null if no trade (not forward-filled)
volume, notional	Σ qty (ETH), Σ price·qty (USD)
vwap	notional / volume (null if no trade)
n_trades	number of aggregated trades; n_fills = number of underlying fills (Σ last_id – first_id + 1)
buy_volume, sell_volume, buy_notional, sell_notional, buy_trades, sell_trades	split by taker side (buy = is_buyer_maker == false)
trade_size_mean, _median, _max, _p90, _p95	distribution of aggTrade sizes within the bar
buy_volume_ge{k}, sell_volume_ge{k}	taker volume from aggTrades with qty $\geq k$ ETH, $k \in \{1, 5, 10, 25, 50, 100, 250\}$. The thresholds are fixed in advance, so Stage 3 can build trailing large-trade features without re-reading the ticks
last_trade_ts	timestamp of the bar's last trade
n_bad_tick_flags	trades in the bar more than 2 % from the bar's median price (QC flag only)
has_trades	n_trades > 0
book_ts	timestamp of the book snapshot attached as-of: last snapshot stamped strictly before ts + 1min
bid_depth_k, ask_depth_k, bid_notional_k, ask_notional_k (k = 1...5)	that snapshot's band values
book_age_s	(ts + 60 s) – last_change_ts (the last snapshot whose values changed), so a frozen feed ages and becomes stale
book_stale	no snapshot, or book_age_s > 120

Column(s)	Definition
book_short_day	the bar's UTC day has < 90 % of the 2,880 expected snapshots (e.g. 2024-06-12, which has only 2). Flag only; nothing is interpolated
book_day_snapshots	number of snapshots on that day

Other outputs

- book_snapshots/ETHUSD-`book_snapshots.parquet`: cleaned snapshots in wide format (ts + the 20 band columns above, book_frozen = all 20 values equal the previous snapshot, last_change_ts).
- research/ETHUSD-`research.parquet` (Stage 3): the bars above plus 47 features (ethof.features.GROUPS), px / exec_px, and the targets fwd_ret_h, up_h, cls3_h, label_end_h for $h \in \{5, 15, 30, 60\}$.
- funding/ETHUSD-`funding.parquet`: calc_time, funding_interval_hours, last_funding_rate, ts.
- metrics/ETHUSD-`metrics.parquet`: the metrics columns + ts.

Cleaning policy

Category	Action
null / non-finite values, $\text{price} \leq 0$, $\text{qty} \leq 0$, first_trade_id > last_trade_id, timestamp outside the file's UTC day, unparseable side flag	removed (counted)
exact duplicate rows	one copy kept (counted)
same agg_trade_id with different content	first kept (counted)
open interest ≤ 0 , $\pm\text{inf}$ feature ratios	set to missing (counted)
frozen (repeated) book snapshots	kept, but treated as stale: book features are missing
trades far from the minute median, extreme 1-min moves, trade-id sequence gaps, empty minutes, missing/short book days, stale snapshots	flagged, kept

Large price moves are never removed just because they are large. The Stage 2 report lists every 1-minute bar with $|\log \text{return}|$ or $\log(\text{high}/\text{low}) > 3\%$ for review. A move is classed as an error candidate only if it fully reverts in the next bar on thin activity.

D. Robustness report (verbatim)

Each row is a robustness test. Values are copied from the named file.

- **Dev** = development walk-forward out-of-fold results (2024-03 → 2025-09).
- **Holdout** = the untouched holdout under the frozen procedure (2025-10 → 2026-09).
- **BE** = break-even cost, in bps per side.

All holdout experiments listed here were declared in docs/final_methodology.md §9 before the holdout was opened.

#	Test	Result	Source
1	Horizon 5 / 15 / 30 / 60 min (holdout AUC, logit + LightGBM; 15 min includes RF)	0.531-0.537 / 0.534-0.544 / 0.530-0.540 / 0.526-0.537. Predictability at every horizon, no horizon stands out	holdout/horizons.csv, holdout/ablation.csv
2	Horizon (dev, same models)	best mean AUC 0.531 / 0.539 / 0.539 / 0.536	model_summary.csv
3	Model class (holdout, set E)	logit 0.534, RF 0.544, LightGBM 0.542. BE 1.07 / 0.49 / 0.55	holdout/ablation.csv
4	Feature ablation A → E (dev, LightGBM)	0.5375 → 0.5378 → 0.5380 → 0.5391 → 0.5388	model_summary.csv
5	Feature ablation A → E (holdout, RF)	0.5428 → 0.5424 → 0.5427 → 0.5434 → 0.5436	holdout/ablation.csv
6	Feature ablation A → E (holdout, logit)	0.5346 → 0.5336 → 0.5363 → 0.5338 → 0.5337; log loss for D and E (0.6939) is worse than the prior (0.6931)	holdout/ablation.csv
7	Cost scenarios (holdout ML, frozen δ)	gross +67.5 %; low (2.52 bps) -89.8 %; base (5.52) -99.6 %; high (7.02) -99.9 %	holdout/backtest_summary.csv
8	Cost sweep (holdout ML)	+67.5 % at 0 bps; -3.9 % at 0.5; -44.8 % at 1.0. BE 0.49 bps	holdout/cost_sensitivity_ml.csv
9	Signal selectivity (dev, EXPLORATORY, non-binding)	mean gross trade +0.19 bps (all minutes) → +0.76 (conf > 0.05) → +2.96 (conf > 0.10, 1.1 % of minutes). Never above the 11.05 bps round trip	dev_edge_by_confidence_EXPLORATORY.csv
10	Volatility regimes (holdout; tercile cut-offs from 2023-03..2024-02)	AUC low 0.550 / mid 0.547 / high 0.541. Gross sum of bar returns -0.117 / +0.168 / +0.496. Net negative in all three	holdout/regimes.csv
11	Volatility regimes (dev)	AUC 0.546 / 0.535 / 0.536. Gross positive only in high volatility. Net negative in all three	dev_regimes.csv
12	Time stability	monthly AUC > 0.5 in 19/19 dev months (0.520-0.558) and 12/12 holdout months (0.521-0.559). Quarterly IC of 15-min OFI is negative in 11/11 dev quarters	dev_monthly_stability.csv, holdout/monthly_stability.csv, stats_quarterly_rank_ic.csv
13	Confirmatory test of the main dev finding on the holdout	1-min OFI → 5 min: β -0.069 bps per sd, t -2.74, Holm p 0.037 (dev: -0.146, t -9.74). → 15 min: not significant (p 0.23)	holdout/confirmatory_tests.csv
14	Replication on SOLUSDT (frozen settings, RF on set E)	dev AUC 0.529, BE 2.08 bps. Holdout AUC 0.534, gross Sharpe 1.74, BE 0.89 bps, net -87.7 % at base	replication_SOLUSDT/{dev_summary, backtest_summary, summary}
15	Alternative targets (dev, LightGBM set E)	3-class: macro AUC 0.614, but UP-vs-DOWN AUC excluding neutral is 0.546, so most of the macro AUC is volatility, not direction. Regression: RMSE 35.66 vs 35.61 for predicting the mean, IC 0.013, so it fails	model_folds_all.csv
16	Execution-price realism	AUC against the VWAP_{t+1} → VWAP_{t+16} return is 0.548 vs 0.544 close-to-close (holdout). The edge survives the one-bar delay statistically	holdout/summary.json

Multiple testing and data snooping

What was tried. Development used 43 model configurations (experiment_registry.csv: 8 tuning points, 15 ablation configurations, 18 horizon configurations, 2 alternative targets), 5 signal thresholds, 36 univariate signal tests, 16 controlled regressions, 12 decile spreads and 4 event definitions.

How the noise was controlled:

- **Univariate tests.** Holm (family-wise error) and Benjamini-Hochberg (false discovery rate) adjustments are applied within each table. Holm is used for the headline claims because the tests are few and strongly dependent. 1-min OFI reversal survives Holm at every horizon in development. 15-min OFI and most other signals do not.
 - **Selection.** Model class, LightGBM hyper-parameters and δ were each chosen by a single pre-declared criterion on development data, then frozen (commit 71ca5ad, SHA-256 f890b16b...). The holdout was evaluated once for ETH and once for the SOL replication. Both accesses are logged with the unchanged methodology hash (reports/holdout_access_log.jsonl).
 - **Bonferroni on the holdout AUC.** With 43 configurations, a one-sided z of 3.05 is required. The holdout AUC has $z = 19.6$ against a day-block bootstrap SE of 0.0022 (holdout/significance.json).
- Predictability is not a selection artefact.**
- **Deflated-Sharpe hurdle.** With $43 \times 5 = 215$ strategy variants and one year of data, the expected maximum Sharpe of zero-skill variants is ≈ 3.28 . The holdout gross Sharpe of 2.03 (bootstrap CI 0.32–3.75) is below this hurdle. The gross profit would not be convincing even if trading were free, and it is irrelevant at realistic costs.
 - **Not used.** A full White reality check / Hansen SPA over the strategy universe was not run. It would only matter if some strategy were net-profitable, and none is: net Sharpe is negative for every model, feature set and cost scenario tested.

Where we deviated from the original plan, and why

1. **Model results produced before the frozen-feed fix were discarded and re-run.** The fix was triggered by a crash ($\pm\text{inf}$ in features), not by results.
2. **Trades are processed from daily files instead of monthly files.** A single month needed more than 13 GB of RAM. This is a computational change with identical content.
3. **Order-book features use percentage bands, not top-N levels,** because the data provide only bands (Stage 1 decision).
4. **A post-holdout bootstrap and snooping accounting was added.** It is inference on frozen predictions, labelled as such.

E. Interview questions and answers

The answers are based on this project's actual results. The source files are listed in `docs/research_report.md`.

Data

Why Binance futures?

- Binance USDS-M is the highest-volume ETH venue.
- Its public archive has years of trade-level data with the aggressor side given explicitly, plus order-book depth, funding and open interest. All of it is SHA-256 verifiable and free, so anyone can rebuild the dataset.
- Coinbase had no free historical book data, and its API, like Tardis, was unreachable from the research environment.
- A perpetual also allows native shorting, which a long/short signal needs.

Why ETH?

- BTC was excluded by design.
- ETH is the second most liquid crypto asset, with ~1M aggregated trades per day.
- SOL was used for replication.

Why this time period?

- The order-book files have gaps before March 2023; the first unbroken 30-day run starts 2023-02-10.
- So 2023-03-01 → 2026-09-30, with the last 12 months reserved as the holdout before any returns were examined.

What are aggTrades?

- Binance merges the fills of one taker order at one price into one aggregated trade.
- Each record has the price, quantity, the first and last underlying trade ids, the timestamp, and the side flag.
- We read 1,877,097,488 of them. 4.0 M underlying fill ids (0.216 %) are missing from the archive; they are flagged, not repaired.

What is buyer-maker?

- `is_buyer_maker = true` means the buyer's order was resting, so the *seller* was the aggressor.
- Buyer-initiated (taker buy) volume is therefore `is_buyer_maker = false`.
- Because the side is given, no Lee-Ready or tick-rule inference is needed.

What does order-flow imbalance mean?

- OFI = (taker buy volume – taker sell volume) / total volume over a window. It ranges from –1 to +1.
- It measures net demand for immediacy.
- It is strongly tied to the *same-minute* return: $\text{corr} = 0.467$.

What exactly is Binance `bookDepth`?

- Cumulative resting quantity and notional within $\pm 1, 2, 3, 4, 5$ % of the price, snapshotted about every 30 s.
- It is **not** a level-by-level book: there is no best bid/ask, spread or queue. ± 1 % is about 300 ticks.
- The sign convention and the UTC time zone were verified empirically: 94 % of snapshots have $\text{bid-band average} < \text{trade VWAP} < \text{ask-band average}$ with no clock shift, against 64 % with a 1-hour shift.
- We also found that the archive repeats one snapshot for a month (2025-04-16 → 2025-05-19). We detect this by checking whether values change.

Statistics

Why is correlation insufficient?

- Pearson correlations here are around 0.005–0.009. With 1.36 M overlapping observations, naive t-stats are hugely inflated, because overlapping forward returns are autocorrelated by construction.
- The returns are heavy-tailed: 1-min excess kurtosis is 487, so a few minutes dominate Pearson.
- Correlation says nothing about the magnitude relative to costs.

- We therefore used HAC errors, non-overlapping re-estimation, rank statistics, decile spreads with a day-block bootstrap, and controls.

Why walk-forward validation?

- Neighbouring minutes share future price paths and regimes, so random CV leaks the future into training.
- Expanding walk-forward mimics live deployment.
- Purge plus a 1-day embargo stops training labels from overlapping validation labels.

What is look-ahead bias, and how was it tested? Look-ahead bias means using information that was not available at decision time. We tested for it directly:

- Every feature was recomputed on data cut at 12 random times. All 34.8 M values were identical: no feature changes when the future is removed.
- Book snapshots must be stamped strictly before the bar close.
- Funding must be settled before decision time.
- Open interest is lagged at least 5 minutes.

What is data snooping, and how was it handled? Data snooping means finding patterns because many things were tried. We recorded every attempt: 43 model configurations, 5 thresholds and 36 univariate tests.

- Holm and BH adjustments were applied to the tests.
- All choices were frozen before the single holdout evaluation.
- The holdout AUC z-score (19.6) passes a Bonferroni correction over 43 configurations (needs 3.05).
- The gross Sharpe (2.03) does *not* beat the ~ 3.3 expected maximum of 215 random strategies, so we do not claim gross profitability either.

Why can AUC > 0.5 still lose money?

- AUC measures ranking of direction, not magnitude net of costs.
- Our model is right 57 % of the time on the trades it takes, with an average gross gain of 0.70 bps.
- A round trip costs 11 bps.
- AUC 0.544 is statistically certain and economically worthless here.

Machine learning

Why logistic regression?

- It is an interpretable, regularised linear baseline (L2, $C = 0.1$) and checks whether non-linearity adds value.
- Holdout AUC 0.534 vs 0.544 for the random forest.

Why gradient boosting (LightGBM)?

- It is the standard strong learner for tabular, noisy, non-linear data with missing values handled natively.
- A restrained 8-point grid picked the *smallest* model (15 leaves, 200 trees). The larger models had worse validation log loss, a sign of overfitting a weak signal.
- The random forest won the pre-declared log-loss rule by 0.00005, a tie in practice.

Why not deep learning?

- The signal-to-noise ratio is tiny: the best IC is about 0.06.
- Tree models already saturate at $AUC \approx 0.54$, and the bottleneck is cost, not model capacity.
- A sequence model would add risk of overfitting and leakage without addressing why the strategy fails.

How did you prevent leakage?

- Causal features only, built on a complete minute grid.
- Backward-only as-of joins with an extra lag for open interest.
- Purge and embargo between training and validation.
- Preprocessing inside per-fold pipelines.
- Holdout rows are accessible only through a guard that requires the frozen methodology file and logs its hash.
- An automated audit runs all of these checks.

How did you tune hyper-parameters?

- 8 configurations, scored on mean validation log loss across 7 walk-forward folds on development data only.
- The holdout was never used for any choice.

Trading

What is the bid-ask spread here?

- The ETHUSDT perp is one tick (\$0.01) wide 99.5 % of the time, a mean of 0.046 bps measured on 3 days.
- Spread is negligible; fees dominate.

What is slippage?

- The difference between the decision price and the achieved price.
- We execute at the next minute's VWAP and add a further 1 bp at base cost (0.5 bp low, 2 bps high).

What are the transaction costs?

- Taker fee 4.5 bps with the BNB discount (5 bps without), plus slippage and half spread: 5.52 bps per side at base, 11 bps per round trip.

What is funding?

- An 8-hourly payment between longs and shorts that keeps the perpetual near spot.
- We charge the realised rate to the position held at each settlement.
- It was immaterial for a 15-minute strategy (+0.08 % of capital over the holdout year).

How does position sizing work?

- Each signal opens 1/15 of capital for 15 minutes; overlapping positions net out, so $|\text{position}| \leq 1$ and there is no leverage.
- Average exposure was 0.22.

Why does a predictive signal not generate alpha?

- The predictable move is short-horizon mean reversion of about 1–3 bps.
- That is the same order of magnitude as the bounce that liquidity providers capture.
- A taker pays about 11 bps to harvest it. Break-even is 0.49 bps per side.

Research

What was your strongest finding?

- Out-of-sample predictability is real and stable: AUC above 0.5 in every one of 31 out-of-sample months, a holdout CI of [0.539, 0.548], and calibrated probabilities.
- It is almost entirely mean reversion in *price*, with order flow pointing the same way (reversal).

What failed?

- The economic hypothesis failed: no model, feature set, horizon, threshold or cost scenario is net-profitable.
- Order-flow and book features did not add robust incremental value over price.
- The continuation rules (momentum, buy-on-OFI) lose money even before costs.

What surprised you?

1. The order-book archive is frozen for a month while looking complete.
2. Order flow predicts reversal, not continuation.
3. The holdout AUC was slightly higher than in development.

What would you do differently?

- Get L2/L3 data to model passive (maker) execution and touch-level imbalance.
- Use second-level horizons.
- Pre-register an even smaller experiment set, and measure SOL's spread for its cost model.

How would you deploy this? Not as a stand-alone taker strategy. Possible uses:

- an execution-timing overlay, e.g. delaying a planned buy after heavy taker buying;
- an input to a market-making skew, if queue-position data confirm a maker edge.

Before that, a paper-trading phase with real fills.

Why should we trust the backtest?

- Raw data are SHA-256-verified and the pipeline is fully reproducible.
- An automated leakage audit runs, and execution happens one bar after the signal at VWAP.
- Costs are sourced or measured, and funding is included.
- The methodology was frozen and hash-logged before a single holdout evaluation.
- Random, momentum and buy-and-hold baselines were evaluated in the same engine.
- The backtest reports a *loss*. The main way it could be wrong is by being optimistic (VWAP fills, 1-bp slippage), which would make the conclusion stronger.

F. Resume bullets

- Built a reproducible, checksum-verified data pipeline. It processes **1.88 billion** Binance ETH/USD-perpetual trades and **3.7 million** order-book depth snapshots into a 1.89-million-row one-minute research dataset. It detected and handled a month-long frozen order-book feed hidden in the public archive.
- Engineered 47 causal microstructure features (order-flow imbalance, trade imbalance, large-trade flow, percentage-band depth imbalance, funding/OI). I verified them with an automated leakage audit that recomputes features on truncated data (34.8 million values, zero differences). HAC regressions with Holm correction showed that 1-minute order flow predicts a small reversal ($t = -9.7$).
- Ran leakage-controlled walk-forward ML (logistic, random forest, LightGBM) with purge/embargo, feature ablation and a frozen, hash-logged methodology. Achieved an untouched-holdout AUC of **0.544** (95% CI 0.539-0.548), well calibrated and above 0.5 in 12/12 months. Replicated on SOL.
- Designed a cost-aware backtest (next-bar VWAP execution, fees, slippage, funding). It showed that the signal's gross Sharpe of **2.03** has a break-even cost of only **0.49 bps/side**, versus a realistic 5.5 bps, so the result is a rigorous negative: **statistically significant but economically untradable**.

G. Reproduction commands and file map

Reproducibility

```
pip install -r requirements.txt          # exact versions; Python 3.13
pytest -q                                # 30 unit tests (uses the 1-hour official-data sample in
    data_sample/)

python scripts/run_stage2.py              # download + SHA-256 verify + QC + 1-min bars (~25 GB streamed; ~1 GB
    kept)
python scripts/measure_spread.py          # quoted spread from bookTicker sample days
python scripts/build_dataset.py           # features + targets -> data/processed/research/
python scripts/leakage_audit.py           # -> docs/leakage_audit.md
python scripts/run_eda_stats.py           # EDA + statistical tests (development only)
python scripts/run_models.py             # walk-forward grid (development only)
python scripts/run_dev_backtest.py        # selections, out-of-fold backtest, regimes
python scripts/run_interpret.py           # permutation importance / SHAP
python scripts/freeze_methodology.py      # writes docs/final_methodology.md (refuses after the holdout is
    opened)
python scripts/run_holdout.py             # FINAL holdout (guarded, logged)
python scripts/run_stage2.py --symbol SOLUSD --data-dir data/sol && \
python scripts/build_dataset.py --symbol SOLUSD --data-dir data/sol && \
python scripts/run_replication.py && python scripts/run_holdout.py --symbol SOLUSD --data-dir data/sol --tag
    replication --skip-secondary
python scripts/run_significance.py && python scripts/make_results_table.py && python scripts/make_notebooks.py
python scripts/make_pdf_report.py         # end-to-end PDF: reports/ETH_OrderFlow_Alpha_Report.pdf
```

- **Data URLs:** <https://data.binance.vision/data/futures/um/{daily|monthly}/{aggTrades|bookDepth|metrics|fundingRate|bookTicker}/{SYMBOL}/{SYMBOL}-{dataset}-{period}.zip> (+ .CHECKSUM). Downloaded 2026-10-07. Each file's URL, size and SHA-256 are written to data/manifest.csv.
- **Repository contents:** raw and processed data are not committed. The repository holds code, configuration, schemas, QC tables, result tables, figures and a small sample.
- **Seeds:** fixed (model seed 20231; bootstrap seeds in code).
- **Configuration:** config/config.yaml.
- **Schema:** docs/data_schema.md.

Repository layout

config/config.yaml	parameters: periods, split, QC thresholds, costs, grids (pre-registered)
src/ethof/	data.py (download/verify) · preprocessing.py (QC, 1-min bars) · pipeline.py ·
quality.py	
evaluation.py	features.py · targets.py · statistics.py · validation.py (folds, holdout guard)
scripts/	models.py · backtest.py (signals, P&L, costs, funding) · strategy.py ·
notebooks/01..08	one script per stage (see Reproducibility)
docs/	executed narrative notebooks that read pipeline outputs
final_methodology	stage1_data_selection · data_quality_report · data_schema · leakage_audit ·
resume_bullets	research_report · robustness_report · results_table · interview_questions ·
reports/tables, figures	every result table and chart
tests/	unit tests: timestamps, order flow, imbalance, rolling features, targets, signals,
costs, P&L, funding	
data_sample/	1-hour subset of official files for tests